A Concise Course in Category Theory

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Chapter 1

Language

1.1 Categories

A *semicategory* is an entity composed by *objects*, *mappings* between objects and associative *law of compositions* of mappings. So, a semicategory is just an entity in which we can talk about *commutative diagrams*. More precisely, is an entity $\mathbf{C}$ given by the following data:

1. a collection $\mathrm{Ob}(\mathbf{C})$, whose elements corresponds to the *objects* of $\mathbf{C}$ and are represented by X, Y, etc;

2. for any two objects X and Y, another collection $\mathrm{Mor}_{\mathbf{C}}(X;Y)$, whose elements represent the *mappings* (or *morphisms*) between X and Y, being usually denoted like an arrow $f : X \to Y$;

3. for any triple of objects, certain functions

$$\circ : \mathrm{Mor}_{\mathbf{C}}(X;Y) \times \mathrm{Mor}_{\mathbf{C}}(Y;Z) \to \mathrm{Mor}_{\mathbf{C}}(X;Z),$$

taking the role of *law of compositions* and satisfying *associativity condition*. That is, for any quadruple of objects, the below diagram is commutative. Shortly, this means that $h \circ (g \circ f) = (h \circ g) \circ f$.

$$\begin{tikzcd}
\mathrm{Mor}_{\mathbf{C}}(X;Y) \times \mathrm{Mor}_{\mathbf{C}}(Y;Z) \times \mathrm{Mor}_{\mathbf{C}}(Z;W) \arrow[r, "id\times\circ"] \arrow[d, "\circ\times id"'] & \mathrm{Mor}_{\mathbf{C}}(X;Y) \times \mathrm{Mor}_{\mathbf{C}}(Y;W) \arrow[d, "\circ"] \\
\mathrm{Mor}_{\mathbf{C}}(X;Z) \times \mathrm{Mor}_{\mathbf{C}}(Z;W) \arrow[r, "\circ"'] & \mathrm{Mor}_{\mathbf{C}}(X;W)
\end{tikzcd}$$

The first idea is consider a semicategory **Set** having sets as objects, functions as mappings and the usual composition of functions as the laws of compositions. Following the given definition, to build such entity we will need consider the *collection* $\mathrm{Ob}(\mathbf{Set})$ of all sets.

Now, remember the Russell paradox, which say that does not exist the set of all sets. So, to build **Set** we can't use *collection* as synonymous of *set*. The same problem holds if we look for the category of all algebraic entities of a certain type (groups, rings, vector spaces, etc) or the category of all topological spaces, to do *axiomatic category theory* we need make a distinction between *collection* and *sets*. Being done such distinction, the categories whose collections of objects and whose collections of morphisms are really sets are usually called *small*.

$$\begin{array}{c}\textbf{definition of}\\ \textbf{collection}\end{array} \Longrightarrow \begin{array}{c}\textbf{definition of}\\ \textbf{category}\end{array}$$

In general we take *collection* as being synonymous of *class* in the sense of Von Neumann–Bernays–Gödel theory. Indeed, make sense talk about the *class* of all sets, as well the class of all groups, all vector spaces and all topological spaces.

On the other hand, we can do *naive* (*semi*)*category theory*, in which we assume that the concept of category is primitive and, therefore, don't need be defined. More precisely, we can assume that the notion of *collection* is primitive, and use the previous definition as a *naive* idea of what a category is. In this perspective, we are subjected to certain paradox like Russel and the obtained theorems will not more than conjectures.

Warning. At the following, we will do *naive math*. So, for us there are not a division between "small semicategories" and "large semicategories". Indeed, there are simply *semicategories*.

Isomorphisms

In the (naive) given definition, a semicategory is composed by objects, morphisms and associative composition. Usually, we work with little more rich entities, in which we also have *identity morphisms*. This means that for any object X there is a morphism $id_X : X \to X$ such that $id_X \circ f = f$ and $g \circ id_X = g$ every when such compositions make sense. More precisely, this means that we suppose the existence of a function

$$id : \mathrm{Ob}(\mathbf{C}) \to \mathrm{Mor}_{\mathbf{C}}(X; X)$$

which assign to any object X a morphism id_X which makes commutative the following diagrams:

**********DIAGRAMS*********

Semicategories with categories are usually called *categories*. The presence of identities is important to talk about *isomorphisms*.

************TALK ABOUT THE CORRESPONDENCE BETWEEN CATEGORIES AND SEMICATEGORIES (THERE IS NO

Examples

There a few important examples of categories which will be used along the text. Some of them was intuitively present in the first part.

Example 1.1. We have the categories **Grp**, **Rng**, $\mathbf{Mod}_R$, **Top** and **Diff**. Their objects are, respectively, groups, rings, R-modules, topological spaces and differentiable manifolds. Their morphisms are functions which preserve additional structures. So, they are homomorphisms in the case of algebraic structures, continuous functions in the case of topological structure and differentiable maps in the case of differentiable strucuture. In any of such entities, the composition is just the composition of functions and the identity morphism is just the identity function. This make sense because the compositions of homomorphisms produces homomorphism, as well the composition of continuous functions also is continuous, etc.

Example 1.2. In the first part of the text, categories build like the above example was called *concrete*. Here we can give a precise formulation of such concept. We call $\mathbf{D}$ a *subcategory* of $\mathbf{C}$ when any object and any morphism of $\mathbf{D}$ also are in $\mathbf{C}$. Furthermore, we also require that composition and the identities are provided from $\mathbf{C}$. More precisely, a subcategory is build selecting subcollections

$$\mathrm{Ob}(\mathbf{D}) \subseteq \mathrm{Ob}(\mathbf{C}) \quad \text{and} \quad \mathrm{Mor}_{\mathbf{D}}(X;Y) \subseteq \mathrm{Mor}_{\mathbf{C}}(X;Y)$$

and taking the compositions as restrictions of the compositions on $\mathbf{C}$. So, we can say that a *concrete category* is just a subcategory of **Set**. A subcategory in which

$$\mathrm{Mor}_{\mathbf{D}}(X;Y) = \mathrm{Mor}_{\mathbf{C}}(X;Y)$$

is called *full*. We observe that the previous examples are concrete categories which are not full. An example of full category can be obtained from **Grp** looking only for the objects given by *abelian groups* and keeping the mappings unchanged.

Example 1.3. In the introduction, we saw intuitively that not every category is concrete, giving as examples the categories of cobordisms. Now, let's look to this with more careful. For a given $n \in \mathbb{N}$, we will build a non-concrete category $\mathbf{Cob}_n$. Their objects are compact n-manifolds M without boundary. The idea is take a morphism $\Sigma : M \to N$ as being a *cobordism*. That is, a $n+1$-manifold Σ, such that $\partial\Sigma = M \sqcup N$. Now, we also need give the associative compositions and identities. The compositions must be functions

$$\circ : \mathrm{Mor}_{\mathbf{Cob}_n}(M;N) \times \mathrm{Mor}_{\mathbf{Cob}_n}(N;M') \to \mathrm{Mor}_{\mathbf{Cob}_n}(M;M')$$

and the main idea to build them is taking gluing along the boundary. More precisely, the idea is take $\Sigma' \circ \Sigma$ as being the "universal completion" (which, as we will see, is a certain kind of universal construction, called *pushout*) of the below diagram. The problem is that such completion is unique up to isomorphisms. So, for the composition be well defined, we need change our morphisms from manifolds Σ such that $\partial\Sigma = M \sqcup N$ from the class of manifolds such that $\partial\Sigma \simeq M \sqcup N$.

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Constructions

Here we present some constructions involving categories. More precisely, we present some ways to build new categories from others.

Example 1.4 (*product category*)**.** Given two categories $\mathbf{C}$ and $\mathbf{D}$ we can define their *product* as being a new category $\mathbf{C}\times\mathbf{D}$, whose collection of objects, collection of morphisms and compositions are just the cartesian product of the correspondent structures of $\mathbf{C}$ and $\mathbf{D}$. In other words, we define the collection of objects as being

$$\mathrm{Ob}(\mathbf{C} \times \mathbf{D}) = \mathrm{Ob}(\mathbf{C}) \times \mathrm{Ob}(\mathbf{D}).$$

So, an object in the product category is just a pair of objects, being one of each category in the product. Furthermore, for any given objects, we define

$$\mathrm{Mor}_{\mathbf{C}\times\mathbf{D}}(X,X';Y,Y') = \mathrm{Mor}_{\mathbf{C}}(X;Y)\times \mathrm{Mor}_{\mathbf{D}}(X';Y').$$

Thus, just for objects, a morphism in the product category is a pair o morphisms. Finally, given (f,f') and (g,g'), we define their composition as being $(f,f')\circ(g,g') = (f\circ g, f'\circ g')$, of mode that the identity of (X,Y) is just (id_X, id_Y).

Example 1.5 (*arrow category*)**.** Another important construction is the *arrow category* of a given $\mathbf{C}$. This is a category, usually represented by $\mathrm{Arr}(\mathbf{C})$, having the morphisms of $\mathbf{C}$ as objects. More precisely, we take

$$\mathrm{Ob}(\mathrm{Arr}(\mathbf{C})) = \bigcup_{X,Y\in \mathrm{Ob}(\mathbf{C})} \mathrm{Mor}_{\mathbf{C}}(X;Y).$$

So, given two objects f and g, we need a notion of mapping between them, like present at the first diagram below. The idea is take $H : f \Rightarrow g$ as being simply a *commutative square* between f and g, like on the second diagram. ************* Diagram ******** Such commutative squares can be identified with pairs (h,h') such that $h'\circ f = g\circ h$. The composition is given by *pasting diagrams*, as represented below. ************** DIAGRAM**********

Many particular subcategories of $\mathrm{Arr}(\mathbf{C})$ are specially interesting. Indeed, fixed an object $A\in\mathbf{C}$, we can fix our attention only the arrows having A as the target space. This define a subcategory $\mathbf{C}/A$, called the *over category*. Their morphisms are commutative squares (h, id_A) and, therefore, can be identified with *commutative triangles* having A at the vertex, like below.
*************DIAGRAM***************

Analogously, we can talk about the *under category* $A/\mathbf{C}$. Their objects are arrows having A as domain. The morphisms are commutative triangles (id_A, h).

- category of spans and of n-spans.
- category of cospans and of n-cospans.

Example 1.6 (*pointed sets*)**.** Let $*$ any set with exactly one object. We can look to the under category $*/\mathbf{Set}$. An object is just a map $*\to X$, which corresponds to a selection of an element $x_o\in X$. We call such element a *basepoint* in X and write (X,x_o) instead of $*\to X$. With this identification, a map $f:(X,x_o)\to(Y,y_o)$ (that is, a morphism between $*\to X$ and $*\to Y$) is just a correspondence $f:X\to Y$ which *preserve the base point*. That is, which satisfy $f(x_o)=y_o$. Because of this, we usually call $*/\mathbf{Set}$ of *category of pointed sets* and write $\mathbf{Set}_*$ to denote it.

We observe that we also have topological spaces formed by an unique point. So, the same intuition can be applied to the category $*/\mathbf{Top}$, of mode that their objects are identified with based topological spaces. Thus, here we also use the notation $\mathbf{Top}_*$.

Similarly, we also have algebraic structures formed by an unique element: the neutral element 0. Therefore, for any algebraic category $\mathbf{Alg}$, we write $\mathbf{Alg}_0$ for the under category $0/\mathbf{Alg}$. But in these categories, something interesting happens. Indeed, remember that the linearity of the homomorphisms imply that they must preserve the neutral element. So, there is an unique morphism $0\to X$. That is, the unique basepoint is the neutral element. But any map $X\to Y$

preserve such basepoint. Therefore, in the algebraic domain, the pointed and the unpointed categories ($\mathbf{Alg}_0$ and $\mathbf{Alg}$) are just the same. This clarify that *algebraic categories are generally simply if compared to topological categories.*

Now, observe that, in any situation (**Set**, **Top** or **Alg**) the "one element entities" $*$ above are characterized by the fact that for any object X there is an unique morphism $X \to *$: the constant map. Because of this, we call $*$ a *terminal object.* So, we can use such characterization to talk about *pointed categories* $\mathbf{C}_*$ associated to any category $\mathbf{C}$. Indeed, they are precisely the under categories $*/\mathbf{C}$ for $* \in \mathbf{C}$ a terminal object.

Example 1.7 (*quotient categories*)**.** fsdfsd

Example 1.8 (*homotopy categories*)**.** homotopy category of spaces and of chain and cochain complex.

1.2 Functors

Here we present another example of category: *the category of all categories.* It will be denoted by **Cat**. Their objects are, evidently, categories. Now, the importance in the build of **Cat** is that their morphisms will be *mappings between categories.* Remembering that categories are essentially areas of math, this means that we will get a notion of "map between areas of math".

The morphisms $F : \mathbf{C} \to \mathbf{D}$ are called *functors* and defined as follow. Remember that a category is composed by objects, morphisms, composition laws and identities. So, the idea is take a map between tehm as being a rule which preserve any of such structures. That is, the idea is take F as being a rule that assign to any object $X \in \mathbf{C}$ another object $F(X) \in \mathbf{D}$, and to any morphism $f \in \mathbf{C}$ another morphism $F(f) \in \mathbf{D}$, of mode that compositions and identities are preserved.

More precisely, we define a functor between $\mathbf{C}$ and $\mathbf{D}$ as being an entity composed of the following data:

1. a function $F : \mathrm{Ob}(\mathbf{C}) \to \mathrm{Ob}(\mathbf{D})$;
2. for any pair of objects, another function $F : \mathrm{Mor}_{\mathbf{C}}(X; Y) \to \mathrm{Mor}_{\mathbf{D}}(F(X); F(Y))$,

of mode that $F(g \circ f) = F(g) \circ F(f)$ and $F(id_X) = id_{F(X)}$. For example, the first of such properties means that, for any objects, the below diagrams are always commutative.

$$\begin{tikzcd}
\mathrm{Mor}_{\mathbf{C}}(X,Y) \times \mathrm{Mor}_{\mathbf{C}}(Y,Z) \arrow[r, "\circ"] \arrow[d, "F\times F"'] & \mathrm{Mor}_{\mathbf{C}}(X,Z) \arrow[d, "F"] \\
\mathrm{Mor}_{\mathbf{D}}(F(X),F(Y)) \times \mathrm{Mor}_{\mathbf{D}}(F(Y),F(Z)) \arrow[r, "\circ"'] & \mathrm{Mor}_{\mathbf{D}}(F(X),F(Z))
\end{tikzcd}$$

Now, to complete the definition of **Cat** we need give an associative way to compose functor for that any category has an identity functor $id_{\mathbf{C}}$. But how functors are just collections of functions, the idea is take the composition $G \circ F$ as being simply the entity formed by the collection of the compositions of the underlying functions. Such law of composition is evidently associative and, for any category $\mathbf{C}$, the rule $id_{\mathbf{C}} : \mathbf{C} \to \mathbf{C}$ which is the identity function in objects and in morphisms can be used as the required identity functor. This complete the build of $\mathbf{Cat}$.

Remark. Observe that, how functors leaves invariant objects, morphisms and compositions, they preserve the commutative of diagrams. More precisely, this means that, if J is any commutative diagram in $\mathbf{C}$, then, for all functors $F : \mathbf{C} \to \mathbf{D}$, the diagram $F(J)$ in $\mathbf{D}$ is also commutative. So, for example, if the first diagram below is commutative, then the second also is.

****************** DIAGRAMA *************

Examples

Let give some examples of functors.

Example 1.9 (*duality principle*)**.** We have a functor $(-)^{op} : \mathbf{Cat} \to \mathbf{Cat}$, called *opposition functor*, build as follow. Given a category $\mathbf{C}$, the correspondent category $\mathbf{C}^{op}$ is obtained from $\mathbf{C}$ simply "inverting arrows". More precisely, the *opposite category* $\mathbf{C}^{op}$ has the same objects than $\mathbf{C}$ and, for any pair of objects, we put

$$\mathrm{Mor}_{\mathbf{C}^{op}}(X;Y) = \mathrm{Mor}_{\mathbf{C}}(Y;X).$$

So, a morphism $f^{op} : X \to Y$ in $\mathbf{C}^{op}$ is just a morphism $f : Y \to X$ in $\mathbf{C}$. The composition is defined by the rule $g^{op} \circ f^{op} = (f \circ g)^{op}$ and the identities are just the identities of $\mathbf{C}$. Now, we need say how $(-)^{op}$ act on the morphisms of $\mathbf{Cat}$. So, given $F : \mathbf{C} \to \mathbf{D}$, we need build $F^{op} : \mathbf{C}^{op} \to \mathbf{D}^{op}$ and such construction must preserve composition of functors and map $id_{\mathbf{C}}$ into the identity of $\mathbf{C}^{op}$. Putting

$$F^{op}(X) = F(X) \quad \text{and} \quad F^{op}(f) = F(f)^{op},$$

a direct verification show that F^{op} is really a functor. The preservation of compositions and identities by $(-)^{op}$ also is simple to verify.

Now, remember that functors preserve commutative diagrams. So, the existence of the functor $(-)^{op}$ say that *any construction (or result) on a category $\mathbf{C}$ described only by commutative diagrams is also valid in the opposite category $\mathbf{C}^{op}$.* This important fact is usually called the *duality principle in category theory.*

Remark. In the literature, a functor defined on an opposite category (or, equivalently, taking values in an opposite category) is usually called *contravariant.* In this text, we will adopt such convention.

A simple example of what duality principle works is the following:

Example 1.10 (*initial objects*)**.**

Let see that each construction with categories present previous is, in truth, a *functorial construction.* This means that they extend to functors $\mathbf{Cat} \to \mathbf{Cat}$.

Example 1.11 (*product functor*)**.** First, let see that the product of categories defines a real functor $\times : \mathbf{Cat} \times \mathbf{Cat} \to \mathbf{Cat}$. To do this, given two functors, we need build a notion of product between them, which must preserve compositions and identities. We define

$$(F \times G)(X, Y) = (F(X), G(Y)) \quad \text{and} \quad (F \times G)(f, g) = (F(f), G(f)).$$

Remark. A functor defined on a product of categories is usually called a *bifunctor*. So, in the previous example, what was proved is that the product of categories is, in fact, a bifunctor.

Example 1.12 (*arrow functor*)**.** Now, let prove that the construction of the arrow category also is functorial. More precisely, let build a functor $\mathrm{Arr} : \mathbf{Cat} \to \mathbf{Cat}$ such that to any category assign their arrow category. So, given $F : \mathbf{C} \to \mathbf{D}$ we need define a new functor $\mathrm{Arr}(F)$ between the correspondent arrow categories, which must satisfy

$$\mathrm{Arr}(G \circ F) = \mathrm{Arr}(G) \circ \mathrm{Arr}(F) \quad \text{and} \quad \mathrm{Arr}(id_{\mathbf{C}}) = id_{\mathrm{Arr}(\mathbf{C})}.$$

The objects of the arrow category are just the morphisms f of $\mathbf{C}$. So, $\mathrm{Arr}(F)$ act naturally in objects by $\mathrm{Arr}(F)(f) = F(f)$. On the other hand, the morphisms in the arrow category are commutative squares. But functors preserve commutative diagrams, of mode that, if $H : f \Rightarrow g$ is a morphism in $\mathrm{Arr}(\mathbf{C})$, we can put $\mathrm{Arr}(F)(H) = f(H)$.

Example 1.13 (*over and under*)**.**

To conclude this section, we give an example of semicategory which is not a category and discuss an extension of such example to give a real category.

Example 1.14 (*semigroups as semicategories*)**.** fsd

- because of the example, semicategories are also called *semigroupoids*. This is example in which the existence of neutral element in the of endomorphisms of each object is (necessary and) sufficient to turn a semicategory into a category.
- grupoids, categories enriched over algebraic categories........

Universality

Being $\mathbf{Cat}$ a category, we can look to their arrow category $\mathrm{Arr}(\mathbf{Cat})$. Their objects are functors and their morphisms are commutative squares of functors. When fixed some $\mathbf{C}$, we also can look to the over and under categories $\mathbf{Cat}/\mathbf{C}$ and $\mathbf{C}/\mathbf{Cat}$. But here we can build an intermediate class of categories: the *comma categories*. Indeed, they are build taking as objects the elements of the under or over categories and as morphisms the morphisms of the arrow categories: commutative squares.

More precisely, given functors $F, G \in \mathbf{Cat}/\mathbf{C}$, we define F/G as being the category whose................................

Classification

Remember that in any category $\mathbf{C}$ we have a natural notion of equivalence between two objects: the *isomorphism*. The most important problem for $\mathbf{C}$ is the *classification problem*. Indeed, we want identify what are the objects of $\mathbf{C}$ up to isomorphisms. More precisely, observe that the isomorphism relation gives an equivalence relation on the collection $\mathrm{Ob}(\mathbf{C})$. So, we look for a bijection between the correspondent quotient space $\mathrm{Iso}(\mathbf{C})$ and another known set X.

In general, the classification problem is really hard! So, the idea is look for *invariants*. An invariant of $\mathbf{C}$ is a property $\mathcal{P}$ such that, if is valid in some object X, then it is also valid in any other object isomorphic to X. So, to conclude that X and Y are *not isomorphic*, enough give an invariant which is valid in X but not in Y.

Now, the problem is: *how get invariants?* Observe any functor leave isorphisms invariants. More precisely, if f is an isomorphism, then $F(f)$ also is. This follow directly by the fact that fuctors preserve composition and identities. So, *any functor on* $\mathbf{C}$ *defines an invariant of* $\mathbf{C}$. Indeed, the invariant is "*to have image by F isomorphic to $F(X)$*". Therefore, to prove that Y is not isomorphic to X, enough give F such that $F(Y)$ is not isomorphic to $F(X)$.

Observe that, thanks to the preservation of isomorphisms by functors, the rule that assign to any category the set of isomorphism class of their objects is functorial. More precisely, we have a functor $\mathrm{Iso} : \mathbf{C} \to \mathbf{Set}$, where $\mathrm{Iso}(F)$ is given passing $F : \mathrm{Ob}(\mathbf{C}) \to \mathrm{Ob}(\mathbf{D})$ to the quotient by the isomorphism equivalent relation. Therefore, we have the following

Conclusion. When we consider only objects up to isomorphisms, we pass from category theory to set theory. So, classifying is a process which subtract "categorical information".

Computation

Example 1.15 (*automorphism groups*)**.**

- preservation of limits
- exact sequences

1.3 Transformations

Here we will see that the category $\mathbf{Cat}$ have more information than other usual categories. Indeed, we will see that we don't have only objects and morphisms, but also a notion of *morphisms between morphisms*. So, we will give a notion of *mapping between functors*, called *natural transformation*. If in any category we know compose their morphisms in a some way, we will see that we have two compatible ways to compose such transformations: *vertically and horizontally*.

In more precise words, we will see that we have a category $\mathrm{Func}(\mathbf{C};\mathbf{D})$ whose objects are functors between fixed categories, whose morphisms are the notion of natural transformations and whose composition $\bullet$ are the vertical composition. Furthermore, we will see that there are bifunctors

$$\circ : \mathrm{Func}(\mathbf{C};\mathbf{D}) \times \mathrm{Func}(\mathbf{D};\mathbf{D}') \to \mathrm{Func}(\mathbf{C};\mathbf{D}'), \tag{1.1}$$

corresponding to the horizontal composition, which are associative. The fact that they are bifunctor translates exactly the compatibility between the two compositions:

$$(\varphi' \bullet \xi') \circ (\varphi \bullet \xi) = (\varphi' \circ \varphi) \bullet (\xi' \circ \xi). \tag{1.2}$$

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So, like for groupoids, where the collection of morphisms are not only sets but have an additional group structure, the collections of morphisms of **Cat** have the additional structure of category. Furthermore, also like for groupoids, where the compositions preserve the additional group structure (that is, are not only function, but homomorphism of groups), in **Cat** the compositions extends to bifunctors.

Conclusion. If a groupoid is category enriched over **Grp**, so the category **Cat** is enriched over yourself and, therefore, corresponds to a 2-category. As was said, the notion of enrichment (as well n-categories) will be discussed in the next chapter.

Proof

A *natural transformation* between two functors $F, G : \mathbf{C} \to \mathbf{D}$ is a rule $\xi : F \Rightarrow G$ that assign to $X \in \mathbf{C}$ a morphism $\xi_X : F(X) \to G(X)$ which is natural in the following sense: for any morphism f on X, the following diagram commutes:

$$\begin{tikzcd} F(X) \arrow[r, "F(f)"] \arrow[d, "\xi_X"'] & F(Y) \arrow[d, "\xi_Y"] \\ G(X) \arrow[r, "G(f)"'] & G(Y) \end{tikzcd} \qquad \begin{tikzcd} \mathbf{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] & \mathbf{D} \end{tikzcd}$$

Let see that such notion really are the morphisms of a certain $\mathrm{Func}(\mathbf{C};\mathbf{D})$. The composition between two $\xi : F \Rightarrow G$ and $\xi' : G \Rightarrow F'$ is the *vertical composition* $\xi' \bullet \xi$, defined simply by $(\xi' \bullet \xi)_X = \xi'_X \circ \xi_X$, as represented in the below diagram. So, such composition is evidently associative and preserve the identity functors.

********************DIAGRAM*********

Now, for any categories we need build bifunctors like (1.1), corresponding to the horizontal composition. At objects we define $\circ$ as being simply the composition of functors. Let define $\circ$ on morphisms. To do this, observe that given natural transformations $\xi : F \Rightarrow G$ and $\xi' : F' \Rightarrow G'$ between functors $F, F' : \mathbf{C} \to \mathbf{D}$ and $G, G' : \mathbf{D} \to \mathbf{H}$, we need define a new transformation

$$\xi' \circ \xi : F' \circ F \Rightarrow G' \circ G.$$

To define it, observe that, by the naturalness of ξ and ξ', the following diagram is commutative for any X, of mode that the idea is take the horizontal composition $\xi' \circ \xi$ as being simply the rule that assign to any X the diagonal of their correspondent diagram.

$$\begin{tikzcd}
F'(F(X)) \arrow[r, "F'(\xi_X)"] \arrow[d, "\xi'_{F(X)}"'] \arrow[dr, dashed, "(\xi'\circ\xi)_X"] & F'(G(X)) \arrow[d, "\xi'_{G(X)}"] \\
G'(F(X)) \arrow[r, "G'(\xi_X)"'] & G'(G(X))
\end{tikzcd}$$

Now, to conclude that $\circ$ really is a bifunctor, we need prove that it preserve identities and compositions. The preservation of identities follows directly from the definition of $\circ$. Indeed, looking to the above diagram we see that $(id_{F'} \circ id_F)_X = id_{F'(F(X))}$. To the preservation of vertical compositions, we need show (1.2). Observe that, by definition of horizontal composition, the left side is exactly the diagonal of the following diagram:

$$\begin{tikzcd}
F'(F(X)) \arrow[r, "F'(\xi_X)"] \arrow[d, "(\varphi'\bullet\xi')_{F(X)}"'] \arrow[dr, dashed] & F'(H(X)) \arrow[d, "(\varphi'\bullet\xi')_{H(X)}"] \\
H'(F(X)) \arrow[r, "H'((\varphi\bullet\xi)_X)"'] & H'(H(X))
\end{tikzcd}$$

But, by the definition of vertical composition, such diagram is equivalent to the following:

$$\begin{tikzcd}
F'(F(X)) \arrow[r, "F'(\xi_X))"] \arrow[d, "\xi'_{F(X)}"'] \arrow[ddrr, dashed] & F'(G(X)) \arrow[r, "F'(\varphi_X)"] & F'(H(X)) \arrow[d, "\xi'_{H(X)}"] \\
G'(F(X)) \arrow[d, "\varphi'_{F(X)}"'] & & G'(H(X)) \arrow[d, "\varphi'_{H(X)}"] \\
H'(F(X)) \arrow[r, "H'(\xi_X)"'] & H'(G(X)) \arrow[r, "H'(\varphi_X)"'] & H'(H(X))
\end{tikzcd}$$

On the other hand, thanks to the naturalness of ξ, φ, ξ' and φ' we can "complete" the interior of the above diagram, obtaining a new commutative diagram:

$$
\begin{tikzcd}
F'(F(X)) \arrow[r, "F'(\xi_X)"] \arrow[d, "\xi'_{F(X)}"'] \arrow[dr, dashed, "(\xi' \circ \xi)_X"] & F'(G(X)) \arrow[r, "F'(\varphi_X)"] \arrow[d, "\xi'_{G(X)}"] & F'(H(X)) \arrow[d, "\xi'_{H(X)}"] \\
G'(F(X)) \arrow[r, "G'(\xi_X)"] \arrow[d, "\varphi'_{F(X)}"'] & G'(G(X)) \arrow[r, "G'(\varphi_X)"] \arrow[d, "\varphi'_{G(X)}"] \arrow[dr, dashed, "(\varphi' \circ \varphi)_X"] & G'(H(X)) \arrow[d, "\varphi'_{H(X)}"] \\
H'(F(X)) \arrow[r, "H'(\xi_X)"'] & H'(G(X)) \arrow[r, "H'(\varphi_X)"'] & H'(H(X))
\end{tikzcd}
$$

But the diagonal arrow of this diagram is exactly the right side of (1.2). Therefore, by commutativity, the left and right must be equal.

Examples

- some examples of natural transformations.
- associator.
- characteristic classes

Equivalences

Here we present three very useful notions of equivalence between categories. We initiate with the more obvious (and more stronger) of them: being **Cat** a category, we can talk about *isomorphisms between categories*. Indeed, we say that $\mathbf{C}$ and $\mathbf{D}$ are isomorphics when are isomorphics as objects of $\mathbf{C}$. That is, when there are functors $F : \mathbf{C} \to \mathbf{D}$ and $G : \mathbf{D} \to \mathbf{C}$ which satisfy the equalities

$$G \circ F = id_{\mathbf{C}} \quad \text{and} \quad F \circ G = id_{\mathbf{D}}. \tag{1.3}$$

We call F an *isomorphism* and G their *inverse*. For that F be an isomorphism is necessary an sufficient that it be bijective on objects and on morphisms. More precisely, is necessary and sufficient that below maps be bijective.

$$F : \mathrm{Ob}(\mathbf{C}) \to \mathrm{Ob}(\mathbf{D}) \quad \text{and} \quad F : \mathrm{Mor}_{\mathbf{C}}(X; Y) \to \mathrm{Mor}_{\mathbf{C}}(F(X); F(Y)).$$

Now, remember that **Cat** is an entity more rich than an usual category. Indeed, their collection of morphisms also are categories. So, we have a notion of isomorphism between functors. Then we can weaken the condition (1.3) requiring isomorphisms instead of equalities. This give a new notion of equivalence between categories. Indeed, we say that $\mathbf{C}$ and $\mathbf{D}$ are *weak isomorphics* (or simply that they are *equivalents*) when there are functor F and G satisfying

$$G \circ F \simeq id_{\mathbf{C}} \quad \text{and} \quad F \circ G \simeq id_{\mathbf{D}}. \tag{1.4}$$

In this case, we call F a *weak isomorphism* and G their *weak inverse*. By definition, the isomorphisms on the functor category $\mathrm{Func}(\mathbf{C};\mathbf{D})$ are just natural transformations $\xi : F \Rightarrow G$ for that there is $\varphi : G \Rightarrow F$ satisfying $\varphi \bullet \xi = id$ and $\xi \bullet \varphi = id$, where id is the trivial natural transformation. This means simply that for any X, we have $\varphi_X \circ \xi_X = id_X$ and $\xi_X \circ \varphi_X = id_X$. So, an isomorphism between F and G is just a transformation ξ between them such that ξ_X is always an isomorphism. Consequently, for a given F, there is a G satisfying (1.4) exactly when F is weak bijective on objects and bijective on morphisms. That is, when the below maps are bijections

$$\mathrm{Iso}(F) : \mathrm{Iso}(\mathbf{C}) \to \mathrm{Iso}(\mathbf{D}) \quad \text{and} \quad F : \mathrm{Mor}_{\mathbf{C}}(X;Y) \to \mathrm{Mor}_{\mathbf{C}}(F(X);F(Y)).$$

Finally, instead of require the existence of functors F and G and natural isomorphisms like in (1.4), we could require the most weak condition: only the existence of natural transformations $\xi : id_{\mathbf{C}} \Rightarrow G \circ F$ and $\varphi : F \circ G \Rightarrow id_{\mathbf{D}}$. But this would be too weak. So, we add some "weak compatibility" between ξ and φ. To get this compatibility, observe that the transformations ξ and φ induces certain triangles

$$\begin{tikzcd} F \arrow[r, Rightarrow, "F\circ\varphi"] \arrow[dr, Rightarrow, "id_F"'] & F \circ G \circ F \arrow[d, Rightarrow, "\varphi\circ F"] \\ & F \end{tikzcd} \qquad \begin{tikzcd} G \arrow[r, Rightarrow, "\xi\circ G"] \arrow[dr, Rightarrow, "id_G"'] & G \circ F \circ G \arrow[d, Rightarrow, "G\circ\varphi"] \\ & G \end{tikzcd}$$

which, in principle, are not commutative (with respect to the vertical composition). We require exactly such commutativity.

In this case, we say that $\mathbf{C}$ and $\mathbf{D}$ are *adjoint categories* and that F and G are *adjoint functors*. We usually call F a *left adjoint* to G and G a *right adjoint* to F. Furthermore, like for isomorphisms and equivalences of categories, we also have a characterization for adjunctions. Indeed, as we will see, F and G are adjoints exactly when there is a natural isomorphism like the following:

$$\mathrm{Mor}_{\mathbf{D}}(F(-);-) \simeq \mathrm{Mor}_{\mathbf{C}}(-;G(-)).$$

Examples

- examples of isomorphic categories;
- examples of equivalent categories (category of vector bundles and projective modules. Morita Theory. Floklore theorem say simply that the category of two dimensional TQFT is equivalent to the category of commutative finite dimensional frobenius algebras);
- examples of adjoint categories

Weakening

Here we observe that the notions of equivalence and adjunction make sense not only in **Cat**, but in any "category enriched over **Cat**". That is, in any "2-category". Such constructions are particular cases of general ways to *weaken* an usual concept in any 2-category, which we discuss briefly.

The idea is the following: consider an usual concept defined using only commutative diagrams. As any 2-category is also a category, such concept make sense in them. But in 2-categories we can weaken the requirement of commutativity simply replacing the equality of morphisms by other two different conditions:

1. the existence of a 2-isomorphism linking such morphisms. That is, considering $f \simeq g$ instead of $f = g$. Here we talk about "diagrams which are commutative up to 2-isomorphisms";

2. the existence of left and right 2-morphisms satisfying certain universal property. Here the intuition is the following: we look for 2-morphisms $\xi : f \Rightarrow g$ and $\varphi : g \Rightarrow f$ as being simply left and right approximations to the strict commutativity. So, we require that such approximations are the best possible, what is translated in terms of a certain universal property.

For example, if we apply such weakening process in the classical concept of isomorphism inside the 2-category **Cat**, we get exactly the notions of equivalence and adjunction of categories. On the other hand, the characterization given fro them in **Cat** are special properties of such category.

To conclude we observe that the universal right and universal left ways to weaken are strictly related. Indeed, if for any usual category $\mathbf{C}$ we can assign a new category $\mathbf{C}^{op}$ simply inverting the morphisms, as will be discussed later, for any 2-category $\mathbf{C}$ we can two categories: $\mathbf{C}^{op}$ and $\mathbf{C}^{co}$. The first is just the opposite category, obtained inverting 1-morphisms and keeping unchanged the 2-morphisms. The second, of course, is obtained keeping invariant the morphisms and inverting the 2-morphisms. So, the right weakened concepts in $\mathbf{C}$ are just the left weakened concepts in $\mathbf{C}^{co}$.

1.4 Representability

For $\mathbf{C}$ there are some special functors, called *hom-functors*. They are special because they are very simple and because they have many nice properties. Such functors are usually denoted by

$$\mathrm{Mor}_{\mathbf{C}}(X, -) : \mathbf{C} \to \mathbf{Set} \quad \text{and} \quad \mathrm{Mor}_{\mathbf{C}}(-, X) : \mathbf{C}^{op} \to \mathbf{Set},$$

or simply by h^X and h_X, being defined in the following way: in objects, they simply assign to any Y the respective sets of morphisms $X \to Y$ and $Y \to X$. In morphisms, they act respectively by composition and precomposition. That is, given $f : Y \to Z$, then

********* DIAGRAM ***********

Examples of important properties that such functors have are, respectively, the preservation of limits and colimits. So, the hom-functors are very useful to do calculations. Such properties are invariant by natural isomorphisms. So, there is special interest not only in the hom-functors, but in any functor natural isomorphic to them. Such functors are called *representable*. The name following from the fact that, if F is a representable functor, then there is an object X which, up to isomorphisms, is sufficient describes F totally.

So, the natural question is: *given a functor $F : \mathbf{C} \to \mathbf{Set}$, when it is representable by a certain object X?* There is a result, called the *Yoneda lemma*, which identify how many natural transformations there are between hom-functor h^X and F. It states that, for any X, the transformations

$\xi : h^X \Rightarrow F$ are in bijection with the elements of $F(X)$ and such bijection are natural in X. More precisely, it states that there is natural isomorphism φ between the functors

$$N, E : \mathrm{Func}(\mathbf{C}; \mathbf{Set}) \times \mathbf{C} \to \mathbf{Set},$$

where E is simply the evaluation functor. On the other hand, in objects N assign to any pair (F, X) the set of natural transformations between h^X and X. In morphisms, N must be a rule that take any natural transformation $\xi : F \Rightarrow G$ and any function $f : X \to Y$ and assign a map

$$N(\xi, f) : \mathrm{Nat}(h^X; F) \to \mathrm{Nat}(h^Y; G).$$

To define such map, first we observe that the hom-functors h^X are functorial in X. This means that there is a functor $h^- : \mathbf{C}^{op} \to \mathrm{Func}(\mathbf{C}; \mathbf{Set})$ that assign to any X the correspondent hom-functor h^X. Indeed, in morphism such functor take any function $f : X \to Y$ and return the natural transformation $h^f : h^Y \Rightarrow h^X$ such that, for any Z,

$$h^f_Z : \mathrm{Mor}_{\mathbf{C}}(Y; Z) \to \mathrm{Mor}_{\mathbf{C}}(X; Z) \quad \text{is} \quad h^f_Z(g) = g \circ f.$$

Now we define $N(\xi, f)$ putting simply

$$N(\xi, f)(\alpha) = \xi \bullet \alpha \bullet h^f,$$

as represented in the below commutative diagram.

$$\begin{tikzcd}
\mathrm{Mor}_{\mathbf{C}}(Y; Z) \arrow[r, "h^Y(g)"] \arrow[d, "h^f_Z"'] \arrow[ddd, dashed, bend right] & \mathrm{Mor}_{\mathbf{C}}(Y; W) \arrow[d, "h^f_W"] \arrow[ddd, dashed, bend left] \\
\mathrm{Mor}_{\mathbf{C}}(X; Z) \arrow[r, "h^X(g)"] \arrow[d, "\alpha_Z"'] & \mathrm{Mor}_{\mathbf{C}}(X; W) \arrow[d, "\alpha_W"] \\
F(Z) \arrow[r, "F(g)"] \arrow[d, "\xi_Z"'] & F(W) \arrow[d, "\xi_W"] \\
G(Z) \arrow[r, "G(g)"'] & G(W)
\end{tikzcd}$$

Proof

Now we will prove the Yoneda lemma. More precisely, we will prove that the rule φ that to any pair assign the map

$$\varphi_{(F,X)} : \mathrm{Nat}(h^X; F) \to F(X), \quad \text{such that} \quad \varphi_{(F,X)}(\xi_X(id_X)),$$

is the required natural isomorphism. First we will see that such map is bijective and after we will prove it really is a natural transformation.

To build an inverse to $\varphi_{(F,X)}$, observe that for any natural transformation $\alpha : h^X \Rightarrow F$ and any object Z, we have the following commutative diagram.

$$\begin{tikzcd}
\mathrm{Mor}_{\mathbf{C}}(X;Y) \arrow[r, "h^X(f)"] \arrow[d, "\alpha_Y"'] & \mathrm{Mor}_{\mathbf{C}}(X;Z) \arrow[d, "\alpha_Z"] \\
F(Y) \arrow[r, "F(f)"'] & F(Z)
\end{tikzcd}$$

By such commutativity we see that, for $Y = X$, we have

$$F(f)(\alpha_Y(id_Y)) = \alpha_Z(h^X(f)(id_Y)) = \alpha_Z(f),$$

of mode that the transformation α is totally determined by the element $\alpha_Y(id_Y) \in F(Y)$. So, for any given element $u \in F(Y)$ we can define a natural transformation putting simply

$$\alpha_Z(f) = F(f)(u),$$

what give clearly an inverse for $\varphi_{(F,X)}$.

Now we will proof that φ is natural. More precisely, we will show that, for any transformation $\xi : F \Rightarrow G$ and any morphism $f : X \to Y$ the below diagram commutes.

$$\begin{tikzcd}
\mathrm{Nat}(h^X;F) \arrow[r, "N(\xi,f)"] \arrow[d, "\varphi_{F,X}"'] & \mathrm{Nat}(h^Y;G) \arrow[d, "\varphi_{G,Y}"] \\
F(X) \arrow[r, "E(\xi,f)"'] & G(Y)
\end{tikzcd}$$

To prove such commutative, given an arbitrary $\alpha : h^X \Rightarrow F$, let see that

$$[E(\xi, f)](\varphi_{(F,X)}(\alpha)) = \varphi_{(G,Y)}([N(\xi, f)](\alpha)).$$

By definition, the right side writes

$$\begin{aligned}
\varphi_{(G,Y)}([N(\xi, f)](\alpha) &= \varphi_{(G,Y)}(\xi \bullet \alpha \bullet h^f) \\
&= (\xi \bullet \alpha \bullet h^f)_Y(id_Y) \\
&= \xi_Y(\alpha_Y(h^f_Y(id_Y))) \\
&= \xi_Y(\alpha_Y(f)).
\end{aligned}$$

Furthermore, the left side writes

$$\begin{aligned}
[E(\xi,f)](\varphi_{(F,X)}(\alpha)) &= [E(\xi,f)](\alpha_X(id_X)) \\
&= (\xi_Y \circ F(f))(\alpha_X(id_X)) \\
&= \xi_Y(F(f)(\alpha_X(id_X))) \\
&= \xi_Y(\alpha_Y(f)),
\end{aligned}$$

where in the last equality (??) was used.

Applications

Now, we will give some immediate applications of the Yoneda lemma. The first consequence is that any category is equivalent to some category of functors. To get such result, we first make two observations:

1. for any functor $F : \mathbf{C} \to \mathbf{D}$ we can assign a subcategory $F(\mathbf{C}) \subset \mathbf{D}$, called the *image of* F, and defined in an obvious way: their objects and their morphisms are the objects and morphims of $\mathbf{D}$ which are of the form $F(X)$ and $F(f)$ for some object $X \in \mathbf{C}$ and some morphism $f \in \mathbf{C}$;

2. we say that F is an *embedding* when their domain is equivalent to their image. By the characterization of weak isomorphisms, this happens exactly when F is inject on morphisms and inject up to isomorphisms on objects. But, as easily can be checked, if F is assumed only bijective on morphisms, then it also is weak inject on objects and, therefore, is a full embedding.

Example 1.16 (*Yoneda embedding*)**.** Remember that for every category $\mathbf{C}$ we have a natural functor

$$h^- : \mathbf{C}^{op} \to \mathrm{Func}(\mathbf{C}; \mathbf{Set}).$$

For given objects X and Y, the Yoneda lemma say that we have bijections

$$\mathrm{Nat}(h^X; h^Y) \simeq \mathrm{Mor}_{\mathbf{C}}(Y; X).$$

This means exactly that the functor h^- is bijective on morphisms and, therefore, by the above remarks, is a full embedding. Now we observe that, by the duality principle we also have a version of the Yoneda lemma for contravariant functors. It say exactly that, for any category, the functor

$$h_- : \mathbf{C} \to \mathrm{Func}(\mathbf{C}^{op}; \mathbf{Set})$$

is also a full embedding, called the *Yoneda embedding.* Consequently, *any category can be naturally identified with a subcategory of contravariant functors.*

Now, as a second application, we will use the Yoneda lemma to give a more concrete characterization of the adjunction relation between two functors.

Example 1.17 (*adjoints*)**.** Remember that $F : \mathbf{C} \to \mathbf{D}$ and $G : \mathbf{D} \to \mathbf{C}$ are called adjoints when there are

$$\xi : id_{\mathbf{C}} \Rightarrow G \circ F \quad \text{and} \quad \varphi : F \circ G \Rightarrow id_{\mathbf{D}},$$

which are weakly compatible in the sense that certain vertical diagrams involving ξ and φ are commutative. Observe that, for any X and Y we have maps

$$\begin{tikzcd} \mathrm{Mor}_{\mathbf{D}}(F(X);Y) \arrow[r, shift right, "\alpha_{XY}"'] & \mathrm{Mor}_{\mathbf{C}}(X;G(Y)) \arrow[l, shift right, "\beta_{XY}"'] \end{tikzcd}$$

which assign to any $f : F(X) \to Y$ and to any $g : X \to G(Y)$ the correspondent dotted arrows on the below diagrams

$$\begin{tikzcd} G(F(X)) \arrow[r, "G(f)"] & G(Y) \\ X \arrow[u, "\xi_X"] \arrow[ur, dashed, "\alpha_{XY}(f)"'] & \end{tikzcd} \qquad \begin{tikzcd} F(X) \arrow[r, "F(g)"] \arrow[dr, dashed, "\beta_{XY}(g)"'] & F(G(Y)) \arrow[d, "\varphi_Y"] \\ & Y \end{tikzcd}$$

The first map α_{XY} is natural at X because ξ_X is natural at X. But for any Y, the given f is in $h^Y(F(X))$ and therefore, by the Yoneda lemma, it corresponds to a transformation $\alpha : h^Y \Rightarrow h^{F(X)}$, which is totally determined by $\alpha_Y(id_Y) = f$. So, the first map is also natural in the argument Y. Analogously, the second map is natural in any of such arguments. Therefore, its defines natural transformations

$$\begin{tikzcd} \mathrm{Mor}_{\mathbf{D}}(F(-);-) \arrow[r, shift right, "\alpha"'] & \mathrm{Mor}_{\mathbf{C}}(-;G(-)) \arrow[l, shift right, "\beta"'] \end{tikzcd}, \tag{1.5}$$

which we affirm be one the inverse of the another. Indeed, for a given f, we have

$$\begin{aligned} \beta_{XY}(\alpha_{XY}(f)) &= \varphi_Y \circ F(\alpha_{XY}(f)) \\ &= \varphi_Y \circ F(G(f) \circ \xi_X) \\ &= \varphi_Y \circ F(G(f)) \circ F(\xi_X). \end{aligned}$$

Now, observe that, by the naturalness of φ, the left below diagram is commutative for any $f : Z \to Y$. In particular, it is commutative for $Z = F(X)$. But, by the weak compatibility between ξ and φ, we also have the right side commutative diagram. Therefore,

$$\beta_{XY}(\alpha_{XY}(f)) = f \circ id_{F(X)} = f.$$

Similarly, we have $\alpha_{XY} \circ \beta_{XY} = id$. So, α and β really are natural isomorphisms, being one the inverse of the another.

$$\begin{tikzcd} F(G(Z)) \arrow[r, "F(G(f))"] \arrow[d, "\varphi_Z"'] & F(G(Y)) \arrow[d, "\varphi_Y"] \\ Z \arrow[r, "f"'] & Y \end{tikzcd} \qquad \begin{tikzcd} F(X) \arrow[r, "F(\xi_X)"] \arrow[dr, dashed, "id_{F(X)}"'] & F(G(F(X))) \arrow[d, "\varphi_{F(X)}"] \\ & F(X) \end{tikzcd}$$

Fortunately, the reciprocal is also valid. More precisely, for given functors F and G, the existence of natural isomorphisms α and β like in (??) also determines weak compatible transformations $\xi : id_{\mathbf{C}} \Rightarrow G \circ F$ and $\varphi : F \circ G \Rightarrow id_{\mathbf{D}}$. With effect, we put simply

$$\xi_X = \alpha_{XF(X)}(id_{F(X)}) \quad \text{and} \quad \varphi_Y = \beta_{G(Y)Y}(id_{G(Y)}),$$

of mode that the naturalness follow directly from the naturalness of α and β. Furthermore, the weak compatibility follow from the fact that α and β really is one the inverse of the another. So, we conclude that *two functors F and G are adjoints exactly when there is a natural isomorphism*

$$\mathrm{Mor}_{\mathbf{D}}(F(-); -) \simeq \mathrm{Mor}_{\mathbf{C}}(-; G(-)).$$

As consequence of the above characterization, we can identify adjunction as been an abstraction of the process which build algebraic structures freely generated by sets:

Example 1.18 (*freely generated structures*)**.** ....

Motivated by the previous example, we will say that a subcategory $\mathbf{D} \subset \mathbf{C}$ is *freely generated by* $\mathbf{C}$ when the inclusion functor $\imath : \mathbf{D} \hookrightarrow \mathbf{C}$ has a (left) adjoint. We give an important example:

Example 1.19 (*path categories*)**.** A category has objects, morphisms, compositions and identities. Forgetting identities we get semicategories. If we forget also the compositions we get what we call a *quiver*. In this context, an object is usually called a *vertex* and the morphisms between objects are called *arrows linking vertex*. We define a morphism between quivers as being simply a rule which preserve the objects and the compositions. There is an obvious forgetful functor $\imath : \mathbf{Cat} \to \mathbf{Quiv}$, for which we asserts the existence of a left adjoint $P : \mathbf{Quiv} \to \mathbf{Cat}$. Consequently, we can talk about *categories free generated by quivers*. Indeed, for a given quiver Q we build their *path category* $P(Q)$. The objects are just the vertex of Q. Given two vertex $x, y \in P(Q)$, a morphism $f : x \to y$ is not simply an arrow between them, but is a *path* linking such vertex. These paths are finite strings of increasing arrows, like the following:

$$x \longrightarrow x_0 \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_n \longrightarrow y$$

For any two given paths $p : x \to y$ and $q : y \to z$, their compositions is given by concatenation, as present below. The identity of each vertex x is simply the *stationary path on* x, which is composed only by x and none arrow.

$$(x \longrightarrow \cdots \longrightarrow y) \circ (y \longrightarrow \cdots \longrightarrow z) = x \longrightarrow \cdots \longrightarrow y \longrightarrow \cdots \longrightarrow z$$

Now, to see that P is really adjoint to $\imath$, observe that a functor $F : P(Q) \to \mathbf{C}$ is exactly a rule that maps vertex of Q on objects of C, and (because the preservation of compositions) which maps paths on strings of composable morphisms, as below.

$$F(x \longrightarrow \cdots \longrightarrow y) = F(x) \longrightarrow \cdots \longrightarrow F(y)$$

This is equivalent to saying that F preserves vertices and arrows, which are exactly the same conditions that a morphism $Q \to \imath(\mathbf{C})$ must satisfy.

Utility

- we can identify many spaces X as being functors $\mathbb{X}$ (more precisely, certain kinds of *sheaves*, as will be discussed briefly in the next chapter and exhaustively in the chapter ??). By the Yoneda Lemma, geometric structures on such spaces, say given by a tensor T, are then identified as being natural transformations $\mathbb{X} \Rightarrow \mathbb{T}$ taking values in a certain "universal" space $\mathbb{T}$.

Chapter 2

Abstraction

2.1 Extensions

Other problems which can be considered in any category $\mathbf{C}$ are the *lifting problem* and the *extension problem* for a given morphism. Indeed, we say that $f : X \to Y$ has the *lifting property* with respect to another map $g : Z \to Y$ (we also say that g is *projective* with respect to f) when there is the dotted arrow making commutative the first below diagram.

$$\begin{tikzcd} & X \arrow[d, "f"] \\ Z \arrow[ur, dashed, "h"] \arrow[r, "g"'] & Y \end{tikzcd} \qquad \begin{tikzcd} & Z \\ Y \arrow[ur, "g"] \arrow[r, "f"'] & X \arrow[u, dashed, "h"'] \end{tikzcd}$$

Analogously, we say that $f : Y \to X$ has the *extension property* with respect to a morphism $g : Y \to Z$ when there is the dotted arrow in the second above diagram. In such case, we also say that g is *injective* with respect to f. In other words, the extension problems in $\mathbf{C}$ are just the lift problems in $\mathbf{C}^{op}$.

The nomenclatures follows from analogy with topology and algebra:

- in *topology*, given a bundle $P \to Y$ and a map over the base space $g : Z \to Y$, the lifting problem refers to the classical problem to lift g to the total space P. On the other hand, given a subespace $A \subset X$ and a map $g : A \to Z$, the extension problem is about the possibility to extend g to the whole space X;

- in *algebra*, a projective R-module P is exactly one such that every map $P \to Y$ is projective with respect to any surjective $f : X \to Y$. Dually, Q is an injective R-module exactly when any injective map taking values in Q is injective with respect to any other map.

For a given morphism $f : X \to Y$, we observe that it has the the lifting property with respect to any other map g (and in this case we say simply that f has the lifting property) exactly when there is a morphism $s : Y \to X$ such that $f \circ s = id_Y$. We call s a *section* for f. That is sufficient is obvious: if s is a section for f, then $s \circ g$ is a lifting for g with respect to f. The necessity is obtained observing that a section is exactly a lift of the identity morphisms id_Y to X with respect to f.

In a totally dual way, we see that $f : Y \to X$ has the extension property if and only if it has a *retraction*: a map r which is a left inverse to f. That is, such that $r \circ f = id_Y$.

Similarly than the classification problem, functors also are very useful in the lifting and extension problems. Indeed, here they also give *obstructions* to the existence of solutions. More precisely, if f has the lifting (resp. extension) property with respect to g, then, for any functor F, the correspondent map $F(f)$ has the lifting (resp. extension) property with respect to the morphism $F(g)$. This follow directly from the fact that functors preserve commutative diagrams. But functors also preserve identities. Consequently, if f has a section s or a retraction r, then $F(s)$ and $F(r)$ are sections and retraction for $F(f)$.

$$\begin{tikzcd}
 & X \arrow[d, "f"] & & & F(X) \arrow[d, "F(f)"] \\
Z \arrow[ur, dashed, "h"] \arrow[r, "g"'] & Y \arrow[rr, Rightarrow, "F", phantom] & & F(Z) \arrow[ur, dashed, "F(h)"] \arrow[r, "F(g)"'] & F(Y)
\end{tikzcd}$$

Example 2.1 (*Brouwer's retraction theorem*)**.** We illustrate the previous ideas giving a very simple proof of that there is no map $f : \mathbb{S}^{n-1} \to \mathbb{D}^n$ with the extension property. This means that, for any given f, there are to many one $g : \mathbb{S}^{n-1} \to X$ with not extends from the sphere to the whole disk with respect to f. Equivalently, we will prove that there is none f having a retraction. Suppose the contrary. Indeed, suppose that there is $r : \mathbb{D}^n \to \mathbb{S}^{n-1}$ such that $r \circ \imath = id$. So, for any $F : \mathbf{Top} \to \mathbf{C}$ the map $F(r)$ also is a retraction for $F(\imath)$. But, as we will see, there are many functors

$$F : \mathbf{Top} \to \mathbf{Grp} \quad \text{such that} \quad F(\mathbb{S}^n) \simeq \mathbb{Z} \quad \text{and} \quad F(\mathbb{D}^n) \simeq 0.$$

Consequently, the extension for $F(f)$ cannot exist (as can be seen in the below diagram), contradicting the hypothesis and ending the proof.

$$\begin{tikzcd}
 & \mathbb{D}^n \arrow[d, dashed, "r"] & & & 0 \arrow[d, "0"] \\
\mathbb{S}^{n-1} \arrow[ur, "f"] \arrow[r, "id"'] & \mathbb{S}^{n-1} & & \mathbb{Z} \arrow[ur, "0"] \arrow[r, "id"'] & \mathbb{Z}
\end{tikzcd} \qquad \overset{F}{\Longrightarrow}$$

Observation. Was given a prove of the Brouwer's retraction theorem using only commutative diagrams and functors. So, by the duality principle, such result is also valid in $\mathbf{Top}^{op}$. This means that there is no map $f : \mathbb{D}^n \to \mathbb{S}^{n-1}$ having a section. In general, this dual result is not commented in the literature.

Example 2.2 (*axiom of choice as existence of sections*)**.**

Weak Extensions

We can consider more stronger and, in some categories, more relaxed versions of the problems of lifting and extension. For example, a way to make such problems more stronger is requiring *uniqueness* of the lifting/extensions. The resultant problems will have evidently more powerful solutions.

To see this, observe that, if we look f and g as objects in $\mathbf{C}/Y$, the required lifting condition means exactly the existence of a morphism $h : g \Rightarrow f$. On the other hand we look f and g as objects in $Y/\mathbf{C}$, the extending condition require exactly the existence of a morphism $h : f \Rightarrow g$. Therefore, require uniqueness is equivalent to require the existence of an unique morphisms between two objects.

So, we can characterizes the morphisms which have the uniqueness lifting property with respect to any map as being exactly the terminal objects of $\mathbf{C}/Y$. Dually, the morphisms which have uniqueness extension property with respect to any map are just the initial objects of $Y/\mathbf{C}$. But terminal and initial objects are unique up to isomorphisms. So, *when we require uniqueness of the lifting/extensions, we gain uniqueness of the solutions.*

Now, to consider weakened versions of the lifting/extension problems, remember that such problems are described using only commutative diagrams. So, if we consider them inside any 2-category, we have two natural ways to weaken them. Indeed, we can talk about *weak lifting/extension problems* and about *universal left and right lifting/extension problems.*

In the first case, we are looking for maps h which lift or extend up to 2-isomorphisms, in the sense that $h \circ f \simeq g$ or $f \circ h \simeq g$. In these situations, we say that f has the *weak lifting/extension property* with respect to g.

On the other hand, we say that $\imath$ has the *universal left lifting property* with respect to f when there is l together a 2-morphism $\xi : f \Rightarrow \imath \circ l$ which corresponds exactly to the best approximation to the usual lifting problem. This means that if $\xi' : f \Rightarrow \imath \circ l'$ is another approximation, then it fits in ξ in an unique way. More precisely, with ξ and ξ' we form an extension problem, as below. So, we require that, for any ξ' such new problem has an unique solution obtained from a 2-morphism $u : l \Rightarrow l'$.

$$\begin{tikzcd} & \imath \circ l' \\ f \arrow[ur, Rightarrow, "\xi'"] \arrow[r, Rightarrow, "\xi"'] & \imath \circ l \arrow[u, Rightarrow, dashed, "\imath \circ u"'] \end{tikzcd} \qquad \begin{tikzcd} & \imath \circ r \arrow[d, Rightarrow, "\varphi"] \\ \imath \circ r' \arrow[ur, Rightarrow, dashed, "\imath \circ u"] \arrow[r, Rightarrow, "\varphi'"'] & f \end{tikzcd}$$

Similarly, we say that $\imath$ has the *universal right lift property* with respect to f when there is r and a best approximation $\varphi : \imath \circ r \Rightarrow f$. That is, such that if $\varphi' : \imath \circ r' \Rightarrow f$ is any other approximation, then the above lifting problem has an unique solutions providing from a 2-morphisms $u : r' \Rightarrow r$.

In a dual way we can define *universal left and right extension problems.* Also observe that to pass from left to right lifting (and similarly to extension), enough invert the 2-morphisms. So, left lifting/extension in $\mathbf{C}$ are just right lifting/extension in $\mathbf{C}^{co}$.

We observe that, by definition, a left universal solution for the extension problem of f with respect to $\imath$ will be given by a morphism l together a best approximation $\xi : f \Rightarrow l \circ \imath$. This best approximation is such that, for any other $\xi' : f \Rightarrow l' \circ \imath$ there is an unique $u : l \Rightarrow l'$ satisfying $\xi' = (u \circ \imath) \bullet \xi$. So, any of such best approximation induces a correspondence

$$t : 2\mathrm{Mor}_{\mathbf{C}}(f; l' \circ \imath) \to 2\mathrm{Mor}_{\mathbf{C}}(l; l'), \quad \text{such that} \quad t(\xi') = u.$$

By the uniqueness of u, such correspondence is injective. It also is surjective, thanks to the commutativity condition. Indeed, given u we recover ξ' by the relation $\xi' = (u \circ \imath) \bullet \xi$. Therefore, we can characterize the left universal solutions of the extension problem as being given by bijections

$$2\mathrm{Mor}_{\mathbf{C}}(f; l' \circ \imath) \simeq 2\mathrm{Mor}_{\mathbf{C}}(l; l'). \tag{2.1}$$

- another way to characterize such solutions is the following: right and left universal extensions as terminal and initial objects.

Kan Extensions

Remember that **Cat** is a 2-category. Therefore, we can talk about lifting/ extension problems, as well about their weakened versions, for given functors having the same domain. As we will see along this section, the solutions of the universal left/right extension problems can be used to describe many information about the underlying domain/codomain category. Such solutions are called *left/right Kan extension*.

So, explicitly, a *left Kan extension* of a functor $F : \mathbf{C} \to \mathbf{D'}$ which respect to other functor $\imath : \mathbf{C} \to \mathbf{D}$ is another functor $L : \mathbf{D} \to \mathbf{D'}$ together a natural transformation $\xi : F \Rightarrow L \circ \imath$ which is universal in the sense that, if $\xi' : F \Rightarrow L' \circ \imath$ is any other transformation, then there is an unique $u : L \Rightarrow L'$ such that first below diagram is commutative.

$$\begin{tikzcd} & L' \circ \imath \\ F \arrow[ur, Rightarrow, "\xi'"] \arrow[r, Rightarrow, "\xi"'] & L \circ \imath \arrow[u, Rightarrow, dashed, "u \circ \imath"'] \end{tikzcd} \qquad \begin{tikzcd} & R \circ \imath \arrow[d, Rightarrow, "\varphi"] \\ R' \circ \imath \arrow[ur, Rightarrow, dashed, "u \circ i"] \arrow[r, Rightarrow, "\varphi'"'] & F \end{tikzcd}$$

Furthermore, a *right Kan extension* of F with respect to $\imath$ is a functor R together an universal natural transformation $\varphi : R \circ \imath \Rightarrow F$. That is, such that if $\varphi' : R' \circ \imath \Rightarrow F$ is any other transformation, then there is an unique $u : R' \Rightarrow R$ making commutative the second above diagram.

We observe that, being particular examples of right and left universal extensions, the Kan extensions are terminal and initial objects of some category and, therefore, they are unique up to isomorphisms.

Suppose that any F has left Kan extensions with respect to a fixed functor $\imath$. Let see that that, in this case, the rule which assign to any F their left Kan extension is functorial. More precisely, we have a functor

$$\mathrm{Lan}_\imath : \mathrm{Func}(\mathbf{C}; \mathbf{D'}) \to \mathrm{Func}(\mathbf{D}; \mathbf{D'}),$$

usually called the *global left Kan extension with respect to* $\imath$, which in objects is the referred rule. To define it on morphisms, given $\alpha : F \Rightarrow G$ we need build another transformation between the correspondents left Kan extensions of F and G. Remember that such extensions are composed not only by functors L_F and L_G, but also for universal natural transformations

$$\xi^F : F \Rightarrow L_F \circ \imath \quad \text{and} \quad \xi^G : G \Rightarrow L_G \circ \imath.$$

Now, observe that the given α induces a new transformation $\xi^G \bullet \alpha$ between F and $L_G \circ \imath$. But, by the universality of L_F, there is an unique $u : L_F \Rightarrow L_G$, which can be considered as the action of $\mathrm{Lan}_\imath$ on α. The preservation of identities and compositions can be rapidly checked.

Following the duality principle, we see that, if any F has right Kan extensions with respect to a given $\imath$, then we also get a notion of *global right Kan extension with respect to* $\imath$, given by a functor

$$\mathrm{Ran}_\imath : \mathrm{Func}(\mathbf{C}; \mathbf{D'}) \to \mathrm{Func}(\mathbf{D'}; \mathbf{D}).$$

Adjoints

As will be evident along the text, Kan extensions are a very fruitful notion, in the sense that it is enough abstract to cover essentially all categorical concept. Indeed, the first edition in the Mac Lane's classical book [**?**] ends with a section entitle "All concepts are Kan extensions".

Part of such power is provided from the following fact: *unlike what happens with strict lifting/extension problems, weak lifting/extensions problems and universal left/right lifting/extensions in general are not invariant by the action of functors.* Indeed, strict problems involve only commutativity, which are described using morphisms and compositions (that is, 1-categorical information). Functors, being mappings between 1-categories preserve such 1-categorical information and, therefore, the strict problems. On the other hand, weak problems involve also 2-categorical information, which are not necessarily preserved by functors.

For example, as will be explained now, given a functor F, *the failure of preservation of weak problems by F measures exactly when it has an adjoint.* We start giving a precise definition of "preservation of weak problems". Indeed, we say that $G : \mathbf{D'} \to \mathbf{E}$ *preserve left Kan extensions* when for any left Kan extension $\xi : F \Rightarrow L \circ \imath$ of any functor $F : \mathbf{C} \to \mathbf{D'}$ with respect to any $\imath$, the transformation

$$G \circ \xi : G \circ F \Rightarrow (G \circ F) \circ \imath$$

also is a left Kan extension (see the below diagram). So, we will prove that, if G is a left adjoint for any other functor, then it preserve left Kan extensions. A dual proof will imply that right adjoints preserve right Kan extensions.

$$\begin{tikzcd} & \mathbf{E} & \\ & \mathbf{D'} \arrow[u, "G"] & \\ \mathbf{C} \arrow[ur, "F"] \arrow[uur, bend left, "G\circ F"] \arrow[rr, "\imath"'] & & \mathbf{D} \arrow[ul, "L"'] \arrow[uul, bend right, "G\circ L"'] \end{tikzcd}$$

We can similarly talk about 1-morphisms which preserve left/right universal extensions in any 2-category. Indeed,...................... Furtheremore, the proof which will be given can be applied in this more abstract context. So, the general formulation is: *left/right adjoints preserve left/right universal extensions.*

Proof

We start with a preliminary and general fact. In any 2-category $\mathbf{C}$, consider a (non necessarily commutative) diagram like the following:

$$\begin{tikzcd} X \arrow[r, shift left, "f"] \arrow[d, "h"'] & Y \arrow[l, shift left, "g"] \arrow[d, "k"] \\ X' \arrow[r, shift left, "f'"] & Y' \arrow[l, shift left, "g'"] \end{tikzcd}$$

There are two ways to follow it: in the clockwise direction or in the anticlockwise direction. For each of them, we have two ways to approximate the commutativity of the diagram: left and right approximations. We assert that if the horizontal arrows are adjunctions, then the left approximations for the clockwise direction are in bijection with the right approximations for the anticlockwise direction.

More precisely, we assert that, over such conditions, the 2-morphisms $\alpha : f' \circ h \circ g \Rightarrow k$ are in bijection with the 2-morphisms $\beta : h \Rightarrow g' \circ k \circ f$. Given α, define the correspondent β as being given by the composition of the following string:

$$h \xRightarrow{\xi' \circ h} g' \circ f' \circ h \xRightarrow{(g' \circ f' \circ h) \circ \xi} g' \circ f' \circ h \circ g \circ f \xRightarrow{g' \circ \alpha \circ f} g' \circ k \circ f \ ,$$

where ξ and ξ' are the units of the supposed adjunctions $f \rightleftharpoons g$ and $f' \rightleftharpoons g'$. Reciprocally, given any β, using the counits φ and φ', we can define the correspondent α as the following composition:

$$f' \circ h \circ g \xRightarrow{f' \circ \beta \circ g} f' \circ g' \circ k \circ f \circ g \xRightarrow{\varphi' \circ (k \circ f \circ g)} k \circ f \circ g \xRightarrow{k \circ \varphi} k$$

That the previous correspondences are really one the inverse of another follow from the weak compatibility between units and counits.

Now we are ready to prove that left adjoints preserve left universal solutions of extension problems. Indeed, let $f' : X' \to Y'$ be a morphism and suppose that it is the left adjoint of another g'. Consider arbitrary maps $l : Z \to X'$ and $k : Z \to Y$. So, we have the first below diagram, which is equivalent to the second.

$$\begin{tikzcd}
Z \arrow[r, shift left, "id"] \arrow[d, "l"'] & Z \arrow[l, shift left, "id"] \arrow[d, "k"] \\
X' \arrow[r, shift left, "f'"] & Y' \arrow[l, shift left, "g'"]
\end{tikzcd}
\qquad
\begin{tikzcd}
Y' \arrow[d, shift left, "g'"] & \\
X' \arrow[u, shift left, "f'"] & \\
& Z \arrow[ul, "l"] \arrow[uul, bend right, "k"']
\end{tikzcd}
\qquad
\begin{tikzcd}
& Y' \arrow[d, shift left, "g'"] & \\
& X' \arrow[u, shift left, "f'"] & \\
X \arrow[ur, "f"] \arrow[uur, bend left, "f' \circ f"] \arrow[rr, "\imath"'] & & Z \arrow[ul, "l"'] \arrow[uul, bend right, "k"']
\end{tikzcd}$$

Therefore, by the previous, we have bijections

$$2\mathrm{Mor}_{\mathbf{C}}(f' \circ l; k) \simeq 2\mathrm{Mor}_{\mathbf{C}}(l; g' \circ k). \tag{2.2}$$

Consider the particular case in which l is the left universal solution to the extension problem of f with respect to $\imath$. So we can "complete" the second diagram, getting the third. Furthermore, we have

$$2\mathrm{Mor}_{\mathbf{C}}(l; g' \circ k) \simeq 2\mathrm{Mor}_{\mathbf{C}}(f; g' \circ k \circ \imath) \simeq 2\mathrm{Mor}_{\mathbf{C}}(f' \circ f; k \circ \imath),$$

where in the first step was used (2.1) for $l' = g' \circ k$ and in the second step was applied (2.2) with l replaced by f and k replaced by $k \circ \imath$. Thus, for any k, the map $f' \circ f$ satisfy

$$2\mathrm{Mor}_{\mathbf{C}}(f' \circ f; k \circ \imath) \simeq 2\mathrm{Mor}_{\mathbf{C}}(f' \circ l; k)$$

and, therefore, by the characterization (2.1), is the left universal solution to the extension problem to $f' \circ l$ with respect to $\imath$. This means that f' really preserve any left universal solution l, ending the proof.

2.2 Limits

In this section we will study the most simple class of Kan extensions: the *limits* and *colimits*. In the next sections, we will see that such extensions are not only the most simple, but also the most important. More precisely, first we will see that certain limits (called *equalizers* and *products*) determines all the others. In the sequence (in the next section) we will prove that limits determines all Kan extensions.

The most simple nontrivial category is $\mathbf{1}$, which have only one object $*$ and whose unique morphism is the identity id_*. So, for any given category $\mathbf{C}$ the most simple functors defined on it are ones which takes values in $\mathbf{1}$. Indeed, $\mathbf{1}$ is a terminal object for $\mathbf{Cat}$ and, therefore, there is an unique of such functors, which will be denoted by $*$.

Consequently, the most elementary Kan extensions in $\mathbf{C}$ are ones involving extensions of $\imath$. So, for any given $F : \mathbf{C} \to \mathbf{D}$, we are looking for left and right Kan extension of F which respect to $\imath$. When exists, they are respectively called the *limit* and the *colimit* of the functor F.

To clarify what limits are, observe that they are composed by a functor $L : \mathbf{1} \to \mathbf{D}$ together an universal natural transformation $\xi : F \Rightarrow L \circ *$. The functor $*$ is constant, of mode that the composition $L \circ *$ also is, being identified with an object of $\mathbf{D}$ which will be denoted by $\lim F$. So, the naturalness of ξ is translated in the below diagrams, called *cones* of F with *vertex* on X_o.

***********DIAGRAM*********

Any other transformation $\xi' : F \Rightarrow L' \circ *$ will have a diagram like the above, but with domain X_o changed by another object X'_o (see the second above diagram). So, for any given ξ', the universality of ξ can be translated in terms of the existence of an unique morphism $u : X'_o \to X_o$ making commutative the first below diagram. Such morphism is a certain kind of "collapse" of the cone with vertex in X'_o in the cone with vertex of X_o. So, succinctly, the limit of F can be see as a terminal object in the category $\mathrm{Cone}(F)$ whose objects are cones and whose morphisms are collapsing maps.

********DIAGRAM**********

Now, remember that right and left extensions are related by inverting 2-morphisms. But here, the 2-morphisms are between arbitrary functors $F : \mathbf{C} \to \mathbf{D}$ and a constant functor taking values in $\mathbf{D}$. So, such transformation can be identified with commutative diagrams (cones) in $\mathbf{D}$. Thus, the description colimits of F are obtained simply inverting the arrow of the cones, getting initial *cocones*, as in the above diagram.

A category $\mathbf{D}$ in which any functor $F : \mathbf{C} \to \mathbf{D}$ has a limit is called *complete*. Similarly, if any functor has colimit, we say that the category is *cocomplete*.

Shapes

Limits and colimits are Kan extensions for functors which respect to the constant functor $* : \mathbf{C} \to \mathbf{1}$. So, what determines the "shape" of the cone and cocones diagrams is just the category $\mathbf{C}$. Consequently, for different domain categories we will have different types of limits and colimits.

Remember that there is a category $\mathbf{Quiv}$ of quivers and an inclusion $\imath : \mathbf{Cat} \hookrightarrow \mathbf{Quiv}$, which posses a left adjoint which was denoted by P. This means that we can talk about categories free generated by a given quiver. The important types of limits are ones in which the domain category is generated by a quiver. Such types are not only important, but also simple to view.

Indeed, for such cases, a functor $F : P(Q) \to \mathbf{D}$ is totally determined by their image over the quiver Q. So, essentially, such functors are in bijection with copies of Q inside $\mathbf{D}$, like present in the following diagram:

******** DIAGRAMA *******

Let characterize some limits with different shapes.

Example 2.3 (*products*)**.** Consider, first, the case in which the category is generated by the quiver with two vertex and none arrow linking then. So, a functor $F : \mathbf{C} \to \mathbf{D}$ is represented simply by two objects of $\mathbf{D}$ labelled by the vertex of $\mathbf{C}$, as present below:

$$\begin{tikzcd} 1 & & X_1 \\ & {} \arrow[r, Rightarrow, "F"] & {} \\ 2 & & X_2 \end{tikzcd}$$

Consequently, a cone for F is simply an object X_o with maps $\pi_i : X_o \to X_i$. A limit for F is an universal cone. Universality means that, for any other cone $\pi'_i : X'_o \to X_i$ there is an unique $u : X'_o \to X_o$ such that $\pi_i \circ u = \pi'_i$. In such limit, the vertex is called *binary product* between the objects X_1 and X_2, being denoted simply by $X_1 \times X_2$. The maps π_i are called *canonical projections*.

$$\begin{tikzcd} & & X_1 \\ X'_o \arrow[r, dashed, "u"] \arrow[urr, bend left, "\pi'_1"] \arrow[drr, bend right, "\pi'_2"'] & X_1 \times X_2 \arrow[ur, "\pi_1"'] \arrow[dr, "\pi_2"] & \\ & & X_2 \end{tikzcd}$$

The given characterization is analogously obtained when $\mathbf{C}$ is generated by a quiver with not only two vertex, but with an arbitrary set of vertex and, again, no arrows linking them.

Example 2.4 (*equalisers*)**.** Now, consider the case in which the category is generated by a quiver with two vertex and two arrows linking then. So, a functor $F : \mathbf{C} \to \mathbf{D}$ can be expressed in the following way:

$$\begin{tikzcd} 1 \arrow[d, shift right] \arrow[d, shift left] & & X_1 \arrow[d, shift right, "f"'] \arrow[d, shift left, "g"] \\ 2 & {} \arrow[r, Rightarrow, "F"] & X_2 \end{tikzcd}$$

Therefore, a cone is an object X_o which maps a and b turning commutative the first below diagram. Such commutativity means simply that $f \circ a = a \circ g$. So, we can equivalently express a cone as the first diagram, supplemented with the condition $f \circ a = a \circ g$.

$$\begin{tikzcd} & X_1 \arrow[dd, shift right, "f"'] \arrow[dd, shift left, "g"] \\ X_o \arrow[ur, "a"] \arrow[dr, "b"'] & \\ & X_2 \end{tikzcd} \qquad \begin{tikzcd} X_o \arrow[r, "a"] & X_1 \arrow[r, shift left, "g"] \arrow[r, shift right, "f"'] & X_2 \end{tikzcd}$$

So, a cone of F is universal when, for every other cone $a' : X'_o \to X_1$ satisfying $f \circ a' = a' \circ g$ there is an unique $u : X'_o \to X_o$ such that $u \circ a = a'$.

$$\begin{tikzcd} X'_o \arrow[r, dashed, "u"] \arrow[rr, bend right, "a'"'] & X_o \arrow[r, "a"] & X_1 \arrow[r, shift left, "g"] \arrow[r, shift right, "f"'] & X_2 \end{tikzcd}$$

When such limit exists, their vertex is called *the equalizer* (or sometimes the *difference kernel*) *between f and g,* being denoted by $eq(f, g)$.

Example 2.5 (*pullbacks*)**.** As a final example, let consider $\mathbf{C}$ as being the category generated by the quiver with three vertex and two arrows with the same target. So, a functor $F : \mathbf{C} \to \mathbf{D}$ is identified with similar quiver in $\mathbf{D}$, like the following:..

********* DIAGRAM *********

A direct application of the duality principle give characterization for the colimits of functors defined on the categories discussed above. Indeed, we can analogously talk about dual products (called *coproducts*), dual equalizers (called *coequalizers*) and dual pullbacks (called *pushouts*). Indeed, the coproducts are characterized by the following diagram:

************ DIAGRAMS*******

Furthermore, coequalizers are characterized by

and, finally, pushouts ......

****

To conclude this subsection we observe that, if a category $\mathbf{D}$ has any limits or colimits of certain shape $\mathbf{C}$, then we have respective functors

$$\lim_{\mathbf{C}} : \mathrm{Func}(\mathbf{C}; \mathbf{D}) \to \mathbf{D} \quad \text{and} \quad \mathrm{colim}_{\mathbf{C}} : \mathrm{Func}(\mathbf{C}; \mathbf{D}) \to \mathbf{D}^{op},$$

which correspond to the global left and right Kan extensions. In this context, such functors are simply called *limit functor* and *colimit functor*.

- when the category has pullbacks, $\lim_{\mathbf{C}}$ is defined exactly on $\mathrm{Span}(\mathbf{C})$. Dual observation.

Dependence

We observe that the many types of limits present in the last subsection are not independent. For example, suppose that $\mathbf{C}$ is a category with have binary products and equalizers between arbitrary maps. We will show that such limits is sufficient to build any pullbacks.

So, let $f : X \to Y$ and $g : Z \to Y$ arbitrary morphisms. Using products and equalizers we need build an universal cone involving f and g, representing their pullback. Remember that such cone must be a commutative square having f and g as adjacent maps, like the first diagram.

****

The main idea is build first a (not necessarily commutative) square involving f and g, and after, turn it commutative and universal. Therefore, we need an object W and maps $W \to X$ and $W \to Y$, like in the second diagram. But there are natural maps with the same domain and taking values in X and in Y. Indeed, are the canonical projections of the product $X \times Y$ (which exists by hypothesis).

At this moment, we have used only the products. So, the idea is use the existence of equalizers. We can force the commutativity of the square simply looking to the equalizer between $f \circ \pi_1$ and $g \circ \pi_2$, like present below:

$$
\begin{tikzcd}
eq(f_1, g_2) \arrow[dr, "a"] \arrow[drr, bend left, "\pi_2 \circ a"] \arrow[ddr, bend right, "\pi_1 \circ a"'] & & \\
 & X \times Y \arrow[r, "\pi_2"] \arrow[d, "\pi_1"'] & Y \arrow[d, "g"] \\
 & X \arrow[r, "f"'] & Z
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
eq(f_1, g_2) \arrow[r, "\pi_2 \circ a"] \arrow[d, "\pi_1 \circ a"'] & Y \arrow[d, "g"] \\
X \arrow[r, "f"'] & Z
\end{tikzcd}
$$

We asserts that the commutative square obtained is also universal and, therefore, computes the pullback between f and g................

************** DIAGRAM **********

In the context of colimits, by the duality principle, we can build pushouts from binary products and coequalizers.

Existence

Now, knowing that the different types of limits are not independent, a natural question is: *there are limits which are really fundamental?* More precisely, *there are limits which determines any others?* At the following we will prove that, if we suppose the existence of arbitrary products (instead of only binary products) and equalizers, then we can build not only pullbacks, but also the limit of any functor.

So, let $F : \mathbf{C} \to \mathbf{D}$ be an arbitrary functor. Using products and equalizers we will build an universal cone for F. Remember that such cone is composed by a vertex X_o and maps $\varphi_X : X_o \to F(X)$ such that the first below diagram commutes for any f. If we write $X = d(f)$ and $Y = c(f)$ and add an identity morphism, then the correspondent diagram take the form of the second diagram.

$$
\begin{tikzcd}
 & F(X) \arrow[dd, "F(f)"] \\
X_o \arrow[ur, "\varphi_X"] \arrow[dr, "\varphi_Y"'] & \\
 & F(Y)
\end{tikzcd}
\qquad
\begin{tikzcd}
 & F(d(f)) \arrow[dr, "F(f)"] & \\
X_o \arrow[ur, "\varphi"] \arrow[dr, "\varphi"'] & & F(c(f)) \\
 & F(c(f)) \arrow[ur, "id"'] &
\end{tikzcd}
$$

So, looking to the last diagram, we have arrows taking values in the same object, like for pullbacks. So, the idea is similar to that was used to build pullbacks from binary products and equalizers. Indeed, we can use products to build canonical projections from the same object to $F(d(f))$ and to $F(c(f))$, like in the first below diagram. The composition of such arrows with $F(f)$ and id are parallel arrows, as represented in the second diagram.

$$\begin{tikzcd}
F(d(f)) \arrow[drr, "F(f)"] & & \\
\prod_X F(X) \arrow[u, "\pi_{d(f)}"] \arrow[d, "\pi_{c(f)}"'] & & F(c(f)) \\
F(c(f)) \arrow[urr, "id"'] & &
\end{tikzcd}
\qquad
\begin{tikzcd}
F(d(f)) \arrow[drr, "F(f)"] & & \\
\prod_X F(X) \arrow[u, "\pi_{d(f)}"] \arrow[d, "\pi_{c(f)}"'] \arrow[rr, dashed, shift left] \arrow[rr, dashed, shift right] & & F(c(f)) \\
F(c(f)) \arrow[urr, "id"'] & &
\end{tikzcd}$$

So, taking the equalizer of such parallel arrows, we we expect get a cone. But observe that the vertex of such "cone" will be defined using explicitly f and, therefore, is not well defined. So, the idea is consider all arrows f at the same time. This can be done introducing the product labelled with f in the other part of the above diagrams. More precisely, instead the previous diagrams, we consider the followings, where in the second of them the dotted arrows was obtained by universality of the products:

$$\begin{tikzcd}
F(d(f)) \arrow[r, "F(f)"] & F(c(f)) \\
\prod_X F(X) \arrow[u, "\pi_{d(f)}"] \arrow[d, "\pi_{c(f)}"'] \arrow[ur] \arrow[dr] & \prod_f F(c(f)) \arrow[u, "\pi_f"'] \arrow[d, "\pi_f"] \\
F(c(f)) \arrow[r, "id"'] & F(c(f))
\end{tikzcd}
\qquad
\begin{tikzcd}
F(d(f)) \arrow[r, "F(f)"] & F(c(f)) \\
\prod_X F(X) \arrow[u, "\pi_{d(f)}"] \arrow[d, "\pi_{c(f)}"'] \arrow[ur] \arrow[dr] \arrow[r, dashed, shift left] \arrow[r, dashed, shift right] & \prod_f F(c(f)) \arrow[u, "\pi_f"'] \arrow[d, "\pi_f"] \\
F(c(f)) \arrow[r, "id"'] & F(c(f))
\end{tikzcd}$$

Taking the equalizer between the previous dotted arrow we get a well defined cone, as present below. By the universality of products and equalizers, such cone will be automatically universal, being the required limit.

$$\begin{tikzcd}
 & F(d(f)) \arrow[r, "F(f)"] & F(c(f)) \\
eq(a,b) \arrow[ur] \arrow[r, "e"] \arrow[dr] & \prod_X F(X) \arrow[u, "\pi_{d(f)}"] \arrow[d, "\pi_{c(f)}"'] \arrow[ur] \arrow[dr] \arrow[r, dashed, shift left, "a"] \arrow[r, dashed, shift right, "b"'] & \prod_f F(c(f)) \arrow[u, "\pi_f"'] \arrow[d, "\pi_f"] \\
 & F(c(f)) \arrow[r, "id"'] & F(c(f))
\end{tikzcd}$$

In a completely dual way, we see that *any colimit is determined by arbitrary coproducts and coequalizers.* The universal cone will be the following, where the diagonal arrows inside the square was obtained from universality of coproducts:

$$\begin{tikzcd}
F(d(f)) \arrow[r, "F(f)"] \arrow[d, "\imath_f"'] & F(c(f)) \arrow[d, "\imath_{c(f)}"] \arrow[dr] & \\
\bigoplus_f F(d(f)) \arrow[r, dashed, shift left, "a"] \arrow[r, dashed, shift right, "b"'] \arrow[ur] \arrow[dr] & \bigoplus_X F(X) \arrow[r] & coeq(a,b) \\
F(d(f)) \arrow[r, "id"'] \arrow[u, "\imath_f"] & F(d(f)) \arrow[u, "\imath_{d(f)}"'] \arrow[ur] &
\end{tikzcd}$$

<u>**Complete Embedding**</u>

Here we will proof that any category can be fully embed in some complete and cocomplete category, corresponding to a certain kind of *canonical completion*. More precisely, we will show that, if $\mathbf{D}$ is complete/cocomplete, then $\mathrm{Func}(\mathbf{C};\mathbf{D})$ also is. As consequence, the Yoneda embed will give the required result.

2.3 Examples

Here we will discuss some examples of complete/cocomplete categories. In each case, by the theorem of existence of limits and colimits, we need only build products and equalizers, as well their dual versions. We also give some examples of categories which cannot admit certain limits/colimits.

Example 2.6 (**Set** *is complete/cocomplete*)**.** The first example of complete and cocomplete category is the category of sets. Indeed their products are just the cartesian products, as well their coproducts are the disjoint unions. Given two parallel functions, say $f, g : X \to Y$, to define their equalizer we need a map $\imath : E \to X$ which is universal with respect to the property $f \circ \imath = g \circ \imath$. Such equality motivate us to consider E as being the biggest subset of X in which f and g are equal, and $\imath$ as the inclusion. The universality is immediate. We observe that, if f and g are totally distinct, then $E = \varnothing$. In this case, the map $\imath : \varnothing \to X$ really exist and is clearly universal, because $\varnothing$ is initial object for **Set**. To build the coequalizer between f and g we need a map $\jmath : Y \to C$ universal with respect to $\jmath \circ f = \jmath \circ g$. The first idea is imitate the build of equalizers and consider the biggest subset $C \subset Y$ in which the image of f and g coincide. But in the case in which f and g are totally distinct, we would have $C = \varnothing$. But there are no map $\jmath : Y \to \varnothing$. But remember that quotient maps are universal. So, the new idea is take C as the quotient of Y by the relation which identify $f(x)$ with $g(x)$ and consider $\jmath$ as being the correspondent quotient map.

As a concrete application of the theorem of existence of colimits, let build the colimit of a certain kind of "inductive process" in **Set** using their coproducts and coequalizers. Evidently, we have a dual construction involving products and equalizers.

Example 2.7 (*inductive limits*)**.** Let I be a poset. So we see it as a quiver whose vertex are the elements of I and there is an arrow $i \to j$ if and only if $i \leq j$. Consequently, for any category $\mathbf{D}$, a functor $F : P(I) \to \mathbf{D}$ is just the same that a parametrized family of objects X_i, with $i \in I$, together arrows $X_i \to X_j$, when $i \leq j$. The colimit of F is called the *inductive limit* of the family X_i. To build it in **Set**, we first consider the coproduct of any $F(i) = X_i$ with respect to the collection of indexes $i \in I$ and with respect to the collection of arrows f. The first are simply the set of all x_i, with $i \in I$. The second, on the other hand, is the collection of pairs (x_i, f), with f defined on i. Now, if $f : i \to j$ is an arrow in I, write $F(f) = f_{ij}$. The commutative diagram for the colimit of F is present below. The commutativity of the square say simply that $a(x_i, f) = f_{ij}(x_i)$ and that $b(x_i, f) = x_i$. So, the colimit of F (which is the coequalizer of a and b) is simply the quotient of the collection of all x_i by the relation with identify x_i with any $f_{ij}(x_i)$.

$$\begin{tikzcd}
X_i \arrow[r, "f_ij"] \arrow[d, "\imath_f"'] & X_j \arrow[d, "\imath_j"] \arrow[dr] & \\
\prod_f X_i \arrow[ur] \arrow[dr] \arrow[r, dashed, shift left, "a"] \arrow[r, dashed, shift right, "b"'] & \bigoplus_i X_i \arrow[r] & coeq(a,b) \\
X_i \arrow[u, "\imath_f"] \arrow[r, "id"'] & X_i \arrow[u, "\imath_i"'] \arrow[ur] &
\end{tikzcd}$$

Example 2.8 (**Cat** *is complete/cocomplete*)**.** As a direct consequence of the completeness and cocompleteness of **Set**, we see that **Cat** also has such properties. Indeed, we can take the product between categories as being the category whose classes of objects and morphisms are just the product of the classes of objects and morphisms of the underlying categories. Furthermore, given functors $F, G : \mathbf{C} \to \mathbf{D}$, we define the equalizer between then as being the category $e(F, G)$ whose objects are the equalizer between the restriction of F and G to class of objects. Similarly, the morphisms of such category are the equalizer between the restriction of F and G to classes of morphisms. So, this is the biggest subcategory of $\mathbf{C}$ in which F and G coincide and, therefore, are universal. Consequently, **Cat** is complete. Similarly we prove the existence of colimits.

Example 2.9 (*free categories*)**.** Now, remember that left adjoint preserve left Kan extensions, as well right adjoints preserve right Kan extensions. So, in particular, if a functor has a right adjoint, then it must preserve colimits. Furtheremore, if it has a left adjoint, then it preserve limits. This can be used to get insights about limits and colimits in free categories. Indeed, let $\mathbf{D} \subset \mathbf{C}$ a subcategory and suppose that $\mathbf{C}$ is free generated by $\mathbf{D}$, of mode that the forgetful functor $\imath : \mathbf{D} \hookrightarrow \mathbf{C}$ has a left adjoint $S : \mathbf{C} \to \mathbf{D}$. Therefore, if some limit exists on $\mathbf{D}$ (say with vertex X), then $\imath(X)$ is also the vertex of a limit on $\mathbf{C}$. Furthermore, any colimit on $\mathbf{D}$ must be freely generated by colimit on $\mathbf{C}$.

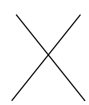
Example 2.10 (*complete/cocomplete concrete categories*)**.** We can apply the previous example to the case of concrete categories. Indeed, if $\mathbf{C} \subset \mathbf{Set}$ is concrete, then it is freely generated by sets. Examples are the algebraic categories **Alg** and **Top**. We use this fact to prove that *such categories also are complete/cocomplete.* Indeed, the products on **Alg** must be the cartesian product endowed with some usual structure with turn the canonical projection linear. Such structure really exists, being given by "pointwise" operations. The equalizer between two parallel linear maps f and g must be the biggest subset in which they coincide, endowed with a natural structure that makes the inclusion linear. Such structure really exists, because the set in question coincide with kernel of $f - g$. So, **Alg** is complete. Similarly, **Top** is complete because we have the product topology and the subspace topology, and because they are the coarsest topologies in which canonical projections and inclusions be continuous. The coproducts of **Alg** and **Top** are just the structures free generated by the coproducts on **Set**.

Incompleteness

How to prove that a given category is *not* complete or cocomplete? Observe that, if $\mathbf{C}$ has some limit/colimit, say of a functor $F : \mathbf{J} \to \mathbf{C}$, then for any embedding $\imath : \mathbf{C} \to \mathbf{D}$, the category $F(\mathbf{C})$ has the limit/colimit of the correspondent $\imath \circ F$. So, to verify that the limit of F does not exist, enough give an embedding $\imath$ such that the limit $F \circ \imath$ exists in $\mathbf{D}$ but not on the subcategory $F(\mathbf{C})$. We can use this observation to proof that the category of differentiable manifold don't has all limits and colimits.

Example 2.11 (**Diff** *is not complete*)**.** Using the previous observation, let see that the category of manifolds is not complete. More precisely, we will show that such category does not have pullbacks. To do this, considering the embed $\mathbf{Diff} \hookrightarrow \mathbf{Set}$, we will prove that certain pullbacks/pushouts involving only manifolds and differentiable maps, when computed on the complete category **Set**, does not admit any differentiable structure. Consider the smooth function $x \mapsto x^2$ between $\mathbb{R}$ and $\mathbb{R}$. So, the first pullback is just the set of all (x, y) such that $x^2 = y^2$ and, therefore, is like an "X", as represented in the third diagram. Thanks to the inverse function theorem, such set clearly cannot admit any smooth structure.

$$\begin{tikzcd} Pb \arrow[r, dashed] \arrow[d, dashed] & \mathbb{R} \arrow[d, "x^2"] \\ \mathbb{R} \arrow[r, "y^2"'] & \mathbb{R} \end{tikzcd} \simeq$$



$$\simeq \begin{tikzcd} Ps & \mathbb{R} \arrow[l, dashed] \\ \mathbb{R} \arrow[u, dashed] & * \arrow[l] \arrow[u] \end{tikzcd}$$

Example 2.12 (**Diff** *is not cocomplete*)**.** Similarly, let see that the category of manifolds does not have pushouts. Indeed, the strategy is identical: we look to the embed $\mathbf{Diff} \hookrightarrow \mathbf{Set}$ and search for pushouts between manifolds such that when computed on the cocomplete category **Set** results in a set with cannot be a manifold. One simple example is given pushouting two maps $* \to \mathbb{R}$, as represented in the last above diagram. Here $*$ is a zero-dimensional manifold consisting of an unique point. Such maps simply select some point inside $\mathbb{R}$, which we denote by x_o and y_o. The pushout is obtaining simply taking the union $\mathbb{R} \sqcup \mathbb{R}$ and then identifying the points x_o and y_o. So, the result also is like an "X" and, therefore, cannot be endowed with a smooth structure.

We finalize observing that the previous examples show that not only **Diff**, but also the full subcategory $\mathbf{Man} \subset \mathbf{Top}$, formed by topological manifolds and continuous maps, is not complete/cocomplete. Indeed, the previous examples was based on the fact that, by the inverse function theorem, two different cartesian spaces cannot be *diffeomorphics.* But such sets not even be *homeomorphics*, of mode that *the space* "X" *cannot admit not even a* C^0 *manifold structure.* This is the content of the invariance of dimension theorem, which can be proved by the following facts

1. the inclusion $\imath : \mathbf{C} \to \mathbf{Top}$ of the full subcategory of compact Hausdorff topological spaces has a left adjoint K, which assign to any space X their so called *Stone-Cech compactification*;
2. as we will see later, there is a sequence of functors $F^i : \mathbf{Top} \to \mathbf{Grp}$ such that we have the identifications
$$F^{i \geq 0}(\mathbb{R}^n) \simeq 0, \quad F^{i \leq n}(\mathbb{S}^n) \simeq 0 \quad \text{and} \quad F^n(\mathbb{S}^n) \simeq \mathbb{Z}$$
for any n. For example, remember that the given proof of Brouwer's retraction theorem was totally based on the existence of a functor like F^n.

To show that the invariance of dimension theorem really follows from the above conditions, suppose the opposite. So, how functors preserve isomorphisms, we must have

$$F^i(K(\mathbb{R}^n)) \simeq F^i(K(\mathbb{R}^m))$$

for any i. But the Stone-Cech compactification of the cartesian space $\mathbb{R}^n$ is homeomorphic to the n-sphere $\mathbb{S}^n$. Consequently, when $n \neq m$ (and supposing $m < n$ without loss of generality) the last condition would imply

$$\mathbb{Z} \simeq F^n(\mathbb{S}^n) \simeq F^n(\mathbb{S}^m) \simeq 0, \quad \text{an absurd.}$$

Another category which are not complete and cocomplete is the homotopy category of topological spaces, as will be discussed now.

Example 2.13 (*incompleteness of $\mathscr{H}$***Top**)**.**

- SEIFERT VAN KAMPEN THEOREM

Smooth Spaces

Remember that the differential geometry is totally build using manifold and differenciable maps between them. That is, differential geometry is totally modeled on the ambient **Diff**. But, as was seen, such ambient is not complete/cocomplete. Consequently, we lost the possibility to do many natural constructions on differential geometry!

There are essentially two ways to avoid the problem of incompleteness of the category of manifolds:

1. *considering specific subcategories* $\mathbf{C} \subset \mathbf{Diff}$ *in which pullback/pushouts exists.* This is the canonical approach when we are doing canonical differential geometry. For example......... transversally........
2. *changing differentiable manifolds for other similar, but less rigid class of spaces.* In the following, we will explain briefly how this can be done. A more detailed exposition will be given in the chapter destined exclusively to geometry.

An way to understand the definition of smooth manifolds is the following: initially, we have the notion of differentiability on the class of cartesian spaces. But, we would like to extend such notion to a more large class of spaces. To do this, we observe that differentiability is a *local concept* (that is, it depends only of the behavior in the neighborhood of any point).

So, if an space M is locally determined by $\mathbb{R}^n$, in the sense that for any $p \in M$ there is a neighborhood U and an open map $\varphi : U \to \mathbb{R}^n$ which is a homeomorphism over their image, then we can transfer the notion of differentiability from $\mathbb{R}^n$ to the neighborhood of any point of X. Now, to extend such notion to the whole space X we need some compatibility in the intersections $U_i \cap U_j$, which can be translated requiring the differentiability of the coordinate change functions $\varphi_j \circ \varphi_i^{-1}$.

In other words, we can think in a manifold as being an space *modelled* over another in which we can talk about differentiability. So, to abstract manifolds we need abstract the *modelling process*. To do this, we observe somethings:

1. in the given interpretation, how we use $\mathbb{R}^n$ to give to M the notion of differentiability, is more natural consider the charts as being defined from some $V_i \subset \mathbb{R}^n$ to M. That is, we look them as embeds $\varphi_i : V_i \to M$;

2. in any manifold, the open sets can be endowed with a natural smooth structure turning the charts $\varphi_i : V_i \to M$ smooth;

3. the whole space M can be build by taking the coproduct of the images of their charts and identifying the intersections.

The last remarks show that *the concept of manifold is totally determined by the structure of the "model space" $\mathbb{R}^n$ and by their open covers*. Indeed, let $\mathrm{Op}(\mathbb{R}^n)$ the category of open subsets $V \subset \mathbb{R}^n$ and smooth maps between them. So we can view a smooth manifold as being simply a space M endowed with a functor $F : \mathrm{Op}(\mathbb{R}^n)^{op} \to \mathbf{Set}$, which assign a set to any open of the cartesian space, making the role of set of smooth maps between V and M and satisfying three conditions:

1. *locality of the differentiability*: for any cover $V_i \hookrightarrow V$ of any open $V \in \mathbb{R}^n$, the first below diagram, obtained from the universality of products is an equaliser;

2. *existence of charts*. More precisely, there are a family of embeds $V_i \hookrightarrow M$;

3. *determination of X by the image of the charts*. This means that the second below diagram is a coequaliser.

So, we can say that the functor F describes the *modelling over* $\mathbb{R}^n$. Now we can use such characterization to give many abstraction of the idea of the modelling process. For example, we can simply discard the role of the charts and look simply to the more fundamental requirement: the locality of differentiability. In this case we can define a *diffeological space* (instead of a manifold) as being simply one entity modeled over $\mathbb{R}^n$ by a functor which satisfy locality.

We can generalize further more! Indeed, observe that in the above notion of locality, we use only the concept of covers of opens $V \in \mathrm{Op}(\mathbb{R}^n)$ and of intersections between elements of such covers. The covers are given by smooth inclusions $V_i \to V$ having the property that if $f : V' \to V$ is any other function, then $f^{-1}(V_i)$ is also a cover for V'. But inverse images are simply pullbacks, as well the intersections. So, we have a categorical description of open covers and then we can talk about *sites*. These are categories $\mathbf{C}$ for which any object $X \in \mathbf{C}$ has a coverage.

Consequently, we can do *modeling over any site* $\mathbf{C}$. This can be done looking to functors $F : \mathbf{C}^{op} \to \mathbf{Set}$ satisfying locality. Such functors are called *sheaves on* $\mathbf{C}$. For example, we can consider a site CSp which contains $\mathrm{Op}(\mathbb{R}^n)$, for any n. It is the category of all cartesian spaces and smooth maps. A cover for a given object is just an open cover for...

- Different models induced by different sites:
  - supersymmetric spaces
  - formal spaces
  - etc.

We can generalize further more! Indeed, we can look to sheaves taking values not on the category of sets, but on an arbitrary category! For example, we can even consider sheaves on **Cat** or any other 2-category. These are more categorical modellings, corresponding to what is called *stacks*. They will be studied in more details in the chapters 10-11. See [??].

- In the next subsection we will see that when we pass from manifolds to diffeological spaces, smooth spaces or other type of "generalized space", we really goes to a category with more properties. More precisely, we will show that the sheaves over a site are equivalent to certain very well behaved categories.

Topos

- sheaves on a site are equivalent to a class of very well behaved categories.
- idea of cohesion.

2.4 Ends

We have proven that the most simple Kan extensions (which are the limits and colimits) can be build from a little piece of information. Indeed, we need only products/coproducts and equalisers/coequalisers. Here we will see that we can use limits and colimits to build any other complicated Kan extension.

So, if a category have products and equalizers, as well their dual versions, then any functor defined on it has Kan extensions with respect to any other functor defined on such category. More precisely, we will prove that, if $\mathbf{D}'$ is cocomplete, then any $F:\mathbf{C}\rightarrow\mathbf{D}'$ has left Kan extensions with respect to any other $\imath:\mathbf{C}\rightarrow\mathbf{D}$. Similarly, if exists colimits, then we can use then to build the right Kan extension of F with respect to $\imath$.

The main idea is the following: given F and $\imath$ as above, we are looking for a functor R and a transformation $\xi : R\circ\imath\Rightarrow F$ satisfying certain universal condition, as represented in the first below diagram. Now, supposing that the category $\mathbf{D}'$ has colimits and remembering that colimits are the most simple left Kan extensions, then, for any of such $\imath$, there is a functor $\overline{R}$ and an universal natural transformation $\overline{\xi}:\overline{R}\circ(1\circ\imath)\rightarrow F$, as present in the second diagram.

********** DIAGRAM *****

We are tented to consider simply $R=\overline{R}\circ 1$ and $\xi=\overline{\xi}$. Such entities are really well defined, but we observe that they are not universal. Indeed, to prove such universality, given any transformation $\xi' : R'\circ\imath\Rightarrow F$, we should obtain an unique $u:R\Rightarrow R'$ satisfying certain commutativity. If R' is of the form $\overline{R}'\circ 1$, then we can get u from the universality of $\overline{R}$. But this is not the general case. This means that the given (wrong) definitions of R and ξ are bad because it not consider some possibilities.

What are we forgetting? Observe that, the condition $R = \overline{R} \circ 1$ make R a constant functor. Consequently, we are ignoring totally the left side of the last diagram. So, the new idea is incorporate the information providing from $\imath$ in the definition of $R(Y)$. Indeed, for a fixed Y, we need incorporate in $R(Y)$ all maps $Y \to \imath(X)$ for any $X \in \mathbf{C}$. We also need that R be determined by limits. So, starting from the construction of limits by products and equalizers, we will add the require additional information.

$$
\begin{tikzcd}
 & F(d(f)) \arrow[r, "F(f)"] & F(c(f)) \\
eq(a,b) \arrow[ur] \arrow[r, "e"] \arrow[dr] & \Pi_X F(X) \arrow[u, "\pi_{d(f)}"] \arrow[d, "\pi_{c(f)}"'] \arrow[ur] \arrow[dr] \arrow[r, dashed, shift left, "a"] \arrow[r, dashed, shift right, "b"'] & \Pi_f F(c(f)) \arrow[u, "\pi_f"'] \arrow[d, "\pi_f"] \\
 & F(c(f)) \arrow[r, "id"'] & F(c(f))
\end{tikzcd}
$$

Remember that the limit of F can be obtained from the above diagram. To define the object $R(Y)$ we need incorporate each map $Y \to \imath(X)$ for all X. So, instead of $\Pi_X F(X)$, the idea is consider their product parameterized by all of such maps. There is, we take

$$\Pi_X \Pi_{\mathrm{Mor}_{\mathbf{D}}(Y;\imath(X))} F(X).$$

Now, if for a fixed Y we define a bifunctor

$$H_y : \mathbf{C}^{op} \times \mathbf{C} \to \mathbf{D}', \quad \text{such that} \quad H_y(X, Z) = \Pi_{\mathrm{Mor}_{\mathbf{D}}(Y;\imath(X))} F(Z),$$

then we see that what we are doing is exactly changing F by $H_y \circ \Delta$, where Δ is the diagonal functor of $\mathbf{C}$, which assign to any X and to any f on $\mathbf{C}$ the correspondent pairs (X, X) and (f, f). So, we expect that we can define $R(Y)$ simply applying such change in the diagram (??), showing that right Kan extensions can really be build from limits.

$$
\begin{tikzcd}
H_y(d(f), d(f)) \arrow[r, dashed] & H_y(c(f), c(f)) \\
\prod_X H_y(X, X) \arrow[u, "\pi_{d(f)}"] \arrow[d, "\pi_{c(f)}"'] \arrow[ur] \arrow[dr] \arrow[r, dotted, shift left] \arrow[r, dotted, shift right] & \prod_f H_y(c(f), c(f)) \arrow[u, "\pi_f"'] \arrow[d, "\pi_f"] \\
H_y(c(f), c(f)) \arrow[r, dashed] & H_y(c(f), c(f))
\end{tikzcd}
$$

Therefore, we are looking to the horizontal arrows in the above diagram. Indeed, if we have them, so we also have the diagonal arrows and, therefore, the central parallel dotted arrows. If we follow exactly our recipe, such arrows would be

$$H_y(f, f) \quad \text{and} \quad H_y(id, id).$$

Observe, however, that this not make sense. Indeed, the functor H_y is contravariant on the first of their arguments. But we observe that, for any f,

$$H_y(id, f) \quad \text{and} \quad H_y(f, id)$$

take values in the same space, of mode that, instead search for a diagram like the last, we can consider the following:

$$\begin{tikzcd}
H_y(d(f), d(f)) \arrow[r, "H_y(id,f)"] & H_y(d(f), c(f)) \\
\Pi_X H_y(X, X) \arrow[u, "\pi_{d(f)}"] \arrow[d, "\pi_{c(f)}"'] \arrow[ur] \arrow[dr] \arrow[r, dashed, shift left, "a"] \arrow[r, dashed, shift right, "b"'] & \Pi_f H_y(d(f), c(f)) \arrow[u, "\pi_f"'] \arrow[d, "\pi_f"'] \\
H_y(c(f), c(f)) \arrow[r, "H_y(f,id)"'] & H_y(d(f), c(f))
\end{tikzcd} \tag{2.3}$$

So, like for limits, we can define $R(Y)$ as being the equalizer between the natural maps a and b. By the functoriality of products and equalizers, such construction is functorial, giving a well defined functor $R : \mathbf{D} \to \mathbf{D}'$. We will prove that such functor really is the right Kan extension of F with respect to $\imath$. More precisely, we will see that, for any other R', it induces a bijection

$$\mathrm{Nat}(R'; R) \simeq \mathrm{Nat}(R' \circ \imath; F).$$

Such construction can be fully dualized. Indeed, dualizing

$$H_y(X, Z) = \Pi_{\mathrm{Mor}_{\mathbf{D}}(Y;\imath(X))} F(Z) \quad \text{we get} \quad H^y(X, Z) = \amalg_{\mathrm{Mor}_{\mathbf{D}}(\imath(X);Y)} F(Z),$$

and dualizing the last diagram we obtain the following:

********** DIAGRAM *******

So, we are led to suspect that not only right Kan extensions are determined by limits, but also left Kan extensions are determined by colimits. More precisely, we suspect that the rule L which to any Y assigns the coequalizer of the above diagram is the left Kan extension of F with respect to $\imath$. As we will see, this is really the case.

Ends and Coends

Before prove the existence of required bijections, let make some observations. Firstly, we observe that the last diagrams make sense not only for H_y and H^y, but also for any bifunctor

$$E : \mathbf{C}^{op} \times \mathbf{C} \to \mathbf{D}'$$

taking values in a complete category. The resultant equalizer (that is, the equalizer of the arrows a and b obtained substituting H_y by E in the last diagram) are usually called *end of E*, being denoted by

$$\int_{\mathbf{C}} E \quad \text{or by} \quad \int_X E(X, X).$$

Furtheremore, the coequalizer of the second diagram is called the *coend* of E and

So, with these new nomenclature and notation, the functors R and L was defined as being the rule which assign to any Y the end of the correspondent H_y and H^y. That is, we have defined

$$R(Y) = \int_X H_y(X,X) \quad \text{and} \quad L(Y) = \int^X H^y(X,X).$$

Example 2.14 (*natural transformation as ends*)**.** Given two functors $F,G : \mathbf{C} \to \mathbf{D}$ we can define a bifunctor

$$\mathrm{Mor}_{\mathbf{Set}}(F(-);G(-)) : \mathbf{C}^{op} \times \mathbf{C} \to \mathbf{Set}.$$

How **Set** is complete, the end of such bifunctor exists. We can compute it with a certain facility. Indeed, remember that the equalizers $eq(a,b)$ is **Set** are just subsets in which $a(x) = b(x)$. Using this in the diagram (??) for the above functor, we see that their end will be the subset in which the first diagram commutes for any f.

$$\begin{tikzcd}
\mathrm{Mor}_{\mathbf{Set}}(F(X),G(X)) \arrow[rd] & \\
\Pi_X \mathrm{Mor}_{\mathbf{Set}}(F(X),G(X)) \arrow[u, "\pi_X"] \arrow[d, "\pi_Y"'] \arrow[r, dashed, shift left] \arrow[r, dashed, shift right] & \mathrm{Mor}_{\mathbf{Set}}(F(X),G(Y)) \\
\mathrm{Mor}_{\mathbf{Set}}(F(Y),G(Y)) \arrow[ru] &
\end{tikzcd}
\qquad
\begin{tikzcd}
F(X) \arrow[r, "F(f)"] \arrow[d, "\xi_X"'] \arrow[rd, dashed, shift left] \arrow[rd, dashed, shift right] & F(Y) \arrow[d, "\xi_Y"] \\
G(X) \arrow[r, "G(f)"'] & G(Y)
\end{tikzcd}$$

So, it is simply the set formed by all families of maps $\xi_X : F(X) \to G(X)$ for that the second diagram is always commutative. But this is just the data which defines a natural transformation. Therefore, we have

$$\int_X \mathrm{Mor}_{\mathbf{D}'}(F(X);G(X)) \simeq \mathrm{Nat}(F;G).$$

- generally, given functors $F,G : \mathbf{C} \to \mathbf{D}$, we can use Yoneda lemma to identify the natural transformations between then only one of those is representable. So, the above example seems a generalization of the Yoneda lemma.

Fubini

In the next subsections, we will give an accurate motivation for the usage of the integral symbols $\int_X$ and $\int^X$, which were used to denote ends and coends. At this moment, we motivate such symbols by a property which remember the Fubini's theorem of the Integration Theory. Indeed, we will show that *the double integrals* $\int_{(X,Y)}$ *can be computed, up to an unique isomorphism, by repeated integrals* $\int_X \int_Y$. So, by duality, the same will holds for coends.

More precisely, let

$$E : (\mathbf{C} \times \mathbf{D})^{op} \times (\mathbf{C} \times \mathbf{D}) \to \mathbf{D}' \tag{2.4}$$

be any functor defined on a product category and taking values in a complete category. So, their end exists, being given by

$$\int_{\mathbf{C}\times\mathbf{D}} E = \int_{(X,Z)} E(X,Z,X,Z).$$

But now, how $(-)^{op}$ is functorial and being the product of categories commutative up equivalences, they induce an equivalence between the categories of functors

$$\mathrm{Func}((\mathbf{C}\times\mathbf{D})^{op}\times(\mathbf{C}\times\mathbf{D});\mathbf{D}')\simeq\mathrm{Func}(\mathbf{C}^{op}\times\mathbf{C}\times\mathbf{D}^{op}\times\mathbf{D};\mathbf{D}'),$$

of mode that any functor like (2.4) can be uniquely identified to another, which will be denoted also by E. Being $\mathbf{D}'$ complete, the projection $E_{\mathbf{C}}$ on $\mathbf{C}$ has ends. That is, for fixed $Z,W\in\mathbf{D}$ we can consider the quantity

$$\int_{\mathbf{C}}E_{\mathbf{C}}=\int_X E(X,X,Z,W).$$

Furthermore, the rule which assign to any $Z,W\in\mathbf{D}$ the correspondent end are functorial, of mode that we can also take their ends

$$\int_{\mathbf{D}}\int_{\mathbf{C}}E=\int_Z\int_X E(X,X,Z,Z).$$

We will prove precisely the categorical analogous of the Fubini's theorem:

$$\int_{\mathbf{C}\times\mathbf{D}}E\simeq\int_{\mathbf{D}}\int_{\mathbf{C}}E\quad\text{or}\quad\int_{(X,Z)}E(X,Z,X,Z)=\int_Z\int_X E(X,X,Z,Z).$$

Before give the prove, we obtain a direct consequence: *the repeated integrals commutes up to isomorphisms.* Indeed, by the Fubini's theorem and by commutativity of the product of categories up to equivalences, we have

$$\int_{\mathbf{D}}\int_{\mathbf{C}}E\simeq\int_{\mathbf{C}\times\mathbf{D}}E\simeq\int_{\mathbf{D}\times\mathbf{C}}E\simeq\int_{\mathbf{C}}\int_{\mathbf{D}}E.$$

The prove of Fubini's theorem require require a very large diagram. Because of this, we move it to the appendix A. See there for details.

Proof

Now we are ready to prove that, for any given $F:\mathbf{C}\to\mathbf{D}'$ and $\imath:\mathbf{C}\to\mathbf{D}$, the functor

$$R(Y)=\int_X H_y(X,X)\tag{2.5}$$

really is the right Kan extension of F with respect to $\imath$. Indeed, for any other functor R',

$$\begin{aligned}
\mathrm{Nat}(R';R) &\simeq \int_Y \mathrm{Mor}_{\mathbf{D}'}(R'(Y);R(Y)) \\
&\simeq \int_Y \mathrm{Mor}_{\mathbf{D}'}(R'(Y);\int_X H_y(X,X)) \\
&\simeq \int_Y \int_X \mathrm{Mor}_{\mathbf{D}'}(R'(Y);H_y(X,X)). \\
&\simeq \int_Y \int_X \mathrm{Mor}_{\mathbf{Set}}(h^Y(\imath(X));h^{R'(Y)}(F(X))) \\
&\simeq \int_X \mathrm{Nat}(h_{\imath(X)};h_{F(X)} \circ R') \\
&\simeq \int_X \mathrm{Mor}_{\mathbf{D}'}(R'(\imath(X));F(X)) \\
&\simeq \mathrm{Nat}(R' \circ \imath;F),
\end{aligned}$$

where is the first step was used the example 2.14, as well in the second was used the definition of R. In the third, was used that the end is an equalizer and that hom-functors preserve such kind of limit. In the forth was applied the definition of H_y and the universal property of products. In the fifth step, the fubini theorem, as well the example 2.14, was used. The Yoneda lemma was applied at the sixth step. Finally, the last bijection was obtained using again the example 2.14.

Mimetizing each of such steps in a dual form, we can prove that left Kan extensions also are determined by the coend of H^y. More precisely, we see that, for any other L',

$$\mathrm{Nat}(L;L') \simeq \mathrm{Nat}(F;L' \circ \imath), \quad \text{where} \quad L(Y) = \int^X H^y(X,X).$$

Feeling

Here we will give a certain feeling about ends and coeds. It is totally based in a similarity between the category of all categories **Cat** and the category of finite dimensional vector spaces $\mathbf{Vec}_{\mathbb{K}}$. Indeed, in such comparison, vector spaces corresponds to categories, as well linear transformations corresponds to functors. Furthermore, the dual space V^* corresponds to the opposite category $\mathbf{C}^{op}$ and we have isomorphisms

$$\mathbf{C}^{op} \simeq \mathbf{C} \quad \text{and} \quad V^* \simeq V.$$

In any of such categories, we also have a well defined operation. Indeed, in $\mathbf{Vec}_{\mathbb{K}}$ we have the tensor product, which can be translated in terms of an "associative" functor

$$\otimes : \mathbf{Vec}_{\mathbb{K}} \times \mathbf{Vec}_{\mathbb{K}} \to \mathbf{Vec}_{\mathbb{K}},$$

for which the field $\mathbb{K}$ behave as an "unity". Similarly, in **Cat** we have the product of categories, which extends to an "associative" functor

$$\times : \mathbf{Cat} \times \mathbf{Cat} \to \mathbf{Cat}$$

having the terminal category $\mathbf{1}$ as "unity". This properties means that, with such operations, both categories are examples of what will be called a *monoidal category*.

Now, in $\mathbf{Vec}_{\mathbb{K}}$ we can talk about *bilinear transformations*. These are just morphisms defined on some tensor product, like $V \otimes W \to Z$. So, the analogous in **Cat** are just the bifunctors. Particularly, we can look to bilinear maps which are contravariant in the first argument and covariant in the second. That is, we can look to maps like $T : V^* \otimes V \to Z$. In correspondence to them we have the bifunctors $\mathbf{C}^{op} \times \mathbf{C} \to \mathbf{D}$, for what we generally can take their end. So, we can ask *what is the linear algebra analogous of* $\int_{\mathbf{C}}$*?*.

Remember somethings about the end:

1. the rule $F \mapsto \int_{\mathbf{C}} F$ is functorial;
2. it take into account only diagonal elements of the form $F(X, X)$.

Consequently, we expect that the linear analogous of the end be given by a linear transformations

$$\mathrm{Mor}_{\mathbf{Vec}_{\mathbb{K}}}(V^* \otimes V; Z) \to Z$$

and that take into account only elements of the form $T(v^*, v)$, where v^* is the dual of v. But T is totally determined by their action on some basis basis $\hat{e}_i$ for the space V. Indeed, we have

$$T(v, v') = \sum_{i,j} v'_i v^j T(\hat{e}^i, \hat{e}_j) = \sum_{i,j} v'_i v^j T^i_j,$$

where v'_i and v^j are just the components of v' and v in the given basis. So, if we look only to the diagonal elements, we are summing over $i = j$. So, we are looking exactly to the operation $T \mapsto \sum_i T^i_i$. Therefore, *the end of a bifunctor is just the categorical analogous of the trace of a bilinear map.*

In this identification, the interchange of two consecutive ends corresponds to the interchange of the order in which the contractions $i \to j$ and $k \to l$ are taken in a tensor of the form T^{ik}_{jl}. In conclusion, we have the following table:

******** TABLE *********

Path Integral

- idea of how see path integral as a kind of Kan extension (as a coend)

Chapter 3

Categorification

Here we are looking to process which allow us to pass from classical concepts to categorical concepts. Any of them will be called a *categorification process*. There are at least two, as will be briefly discussed in the first section. Indeed, they are given by the following strategies:

1. remember that the given concept (being classical) is defined in terms of sets, functions, relations, etc. So, to get a categorical concept we can simple substitute the classical information for categorical information. That is, where in the the given concept we have sets, we substitute them by categories. Where we have functions, we substitute by functors, and so on. This strategy will corresponds to what will be called *internalization in* **Cat**;

2. remember that a (locally small) category $\mathbf{C}$ is composed by a collection of objects, as well *sets of morphisms*, compositions and unities. The compositions are just *functions* between the *cartesian product* of the *sets of morphisms*. So, in the definition of category was used many classical information. Here, the idea is consider entities in which such classical information is substituted by categorical information. That is, we can look to entities formed by a collection of objects, *categories of morphisms*, which compositions given by *bifunctors* between the *product* of the *categories of morphisms*. Such process will be called *enrichment over* **Cat**;

3.1 Internalization

The *internalization process* of a given concept $\mathcal{P}$ consists of two steps:

1. characterization of the concept in terms of purely categorical properties (that is, in terms of diagrams, involving limits, etc);

2. definition of an analogous in any category $\mathbf{H}$ having such properties. We say that the result is $\mathcal{P}$ *internalized in* $\mathbf{H}$ (called the *ambient* of the internalization).

There are many interesting examples. Let see some of them.

Example 3.1 (*monoids*)**.** We consider $\mathcal{P}$ as being the classical concept of monoid. This means that we have a set M together an associative operation $* : M \times M \to M$ and a distinguished neutral element $1 \in M$. This conditions can be translated in the commutativity of the following diagrams:

$$
\begin{tikzcd}
M \times M \times M \arrow[r, "* \times id"] \arrow[d, "id \times *"'] & M \times M \arrow[d, "*"] \\
M \times M \arrow[r, "*"'] & M
\end{tikzcd}
\qquad
\begin{tikzcd}
1 \times M \arrow[r] \arrow[dr, "\pi_1"'] & M \times M \arrow[d, "*"] & M \times 1 \arrow[l] \arrow[dl, "\pi_2"] \\
& M &
\end{tikzcd}
$$

The distinguished element can be seen as a morphism $1 \to M$, where 1 is a terminal object of **Set**. So, the concept of monoid is given by a morphism between a binary product, a morphism from a terminal object for that certain diagrams (involving products between three objects) are commutative. These are purely categorical information, of mode that we can define monoid internal to any **H** in which such information make sense. In particular, *we can define monoid internal to any category with finite products.* For example, we can talk about monoid internal to **Top**. These are simply topological spaces endowed with associative continuous product and a distinguished point. The same can also be done in **Diff** and even in **Cat**. In this last situation, we get what is called a *strict monoidal category.*

Example 3.2 (*commutative monoids and groups*)**.** There is an analogous characterization for commutative monoids and for groups. In the first case, the requirement $x * y = y * x$ can be translated in the commutativity of the first below diagram. Furthermore, the existence of inverses can be translated by the existence of a new function $inv : M \to M$ such that the second below diagram is commutative. Here, $\Delta : M \to M \times M$ is the diagonal map $\Delta(x) = (x, x)$. We observe that such map exists in any category with binary products. Indeed, it is provided by the universality of products, applied to the identity morphisms, as in the third diagram. So, we can internalize commutative monoids and groups (as well commutative groups!) in any category in which we can internalize monoids. For example, the groups internal to **Top**, **Diff** and **Cat** are simply the topological groups, Lie groups and what is would try to call "groupoidal categories". The commutative monoids internalized in **Cat** are usually called *symmetric strict monoidal categories.*

************ DIAGRAMS *************

Example 3.3 (*quivers*)**.** Remember that a quiver Q is composed by a set of objects and a set of morphisms between any two fixed objects. We can join all sets of morphisms in an unique set and say that, equivalently, a quiver Q is simply an entity composed by two sets $\mathrm{Ob}(Q)$ and $\mathrm{Mor}(Q)$, and two arrows d and c, which assign to any f the respective objects $d(f)$ and $c(f)$, corresponding to their domain and codomain. So, to talk about quivers, we only need objects and morphisms. So, *quivers can be internalized in any category.*

The above examples clarify that when we are internalizing in **Cat** a concept initially defined on a concrete category, what we are doing is just replacing objects by categories, functions by functors, distinguished elements by distinguished objects and commutative diagrams between their analogous. Because of this, we usually say that internalize in **Cat** is a way to *categorify.* Furthermore, when a new concept is obtained by such process, we say that it was obtained from *categorification by internalization.*

Non Strictness

- OBS: is very simple give examples of topological monoids, smooth monoids. But the "expected" examples of monoidal categories don't work. Problem is the strictness.
- examples of non strict monoidal categories. Any category with binary products or coproducts is a (non strict) monoidal category.
- smash product.
- monoidal structure in category of spans.
- monoidal structure in cobordism category
- monoidal structure in $\mathbf{C}^n$ with connected sum.
- MONOIDAL FUNCTORS.
- CATEGORY OF MONOIDAL CATEGORIES Str**MonCat**
- Recall that the classical examples of monoidal categories are not strict. The same holds for monoidal functors. This means that, in general the conditions (**??**) holds only up to isomorphisms. Some examples of such kind of monoidal functors are the following:
  1. SEIFERT-VAN KANPEM IMPLY THAT π_1 IS MONOIDAL;
  2. TOPOLOGICAL QUANTUM FIELD THEORIES.
  3. KUNNET THEOREM SAY THAT COHOMOLOGY IS MONOIDAL

Example 3.4 (*equivalent categories with different monoidal structures*)**.** Consider $\mathbf{C}$ with cartesian and cocartesian monoidal structure. $\mathbf{C}$ and $\mathbf{D}$ which are equivalent but are not equivalent as $\mathbf{H}$-internalized categories. More precisely, in it we give two monoidal categories which are equivalent as categories but are not monoidal equivalents. $\mathrm{Span}_n(\mathbf{C}/A)$ and $\mathrm{Span}_n(\mathbf{C}, A)$ can be regarded as see https://ncatlab.org/nlab/show/prequantum+field+theory

Internalized Categories

Here we will see that the internalization process also can be applied to categories! The first step is give a *categorical characterization of the concept of category.* The idea to do this is the following: we have seen that quivers can be easily characterized. A category, on the other hand, can be seen as a quiver added of compositions and identities. So we can say that a category $\mathbf{C}$ is formed not only by two sets and two arrows c, d, but also by other two arrows

$$id : \mathrm{Ob}(\mathbf{C}) \to \mathrm{Mor}(\mathbf{C}) \quad \text{and} \quad \circ : \mathrm{Mor}(\mathbf{C}) \times \mathrm{Mor}(\mathbf{C}) \to \mathrm{Mor}(\mathbf{C}),$$

which will assign to any object their identity morphism, and $\circ$ which take two arrows are return their composition. But observe that in general we can compose only arrows f and g such that f ends when g stars. That is, such that $c(f) = d(g)$. So, in general $\circ$ is defined only in the subset of $\mathrm{Mor}(\mathbf{C}) \times \mathrm{Mor}(\mathbf{C})$ in which $c = d$. But this subset is just the pullback between c and d!

Now, to say that id and $\circ$ really are rules assigning identities and compositions, we need specify some of other things. For example, we need say that $id(X)$ is really an endomorphism of X. Furthermore, we also need say that the composition $g \circ f$ is defined between $d(f)$ and $c(g)$. These information are described in terms of the commutativity of the following diagrams:

$$
\begin{tikzcd}
\mathrm{Ob}(\mathbf{C}) \arrow[r, "id"] \arrow[dr, "id"'] & \mathrm{Mor}(\mathbf{C}) \arrow[d, shift right, "c"'] \arrow[d, shift left, "d"] \\
& \mathrm{Ob}(\mathbf{C})
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathrm{Pb}(d,c) \arrow[dr, "id"] \arrow[drr, bend left, "\circ"] \arrow[ddr, bend right, "\circ"'] & & \\
& \mathrm{Pb}(d,c) \arrow[r] \arrow[d] & \mathrm{Mor}(\mathbf{C}) \arrow[d, "c"] \\
& \mathrm{Mor}(\mathbf{C}) \arrow[r, "d"'] & \mathrm{Ob}(\mathbf{C})
\end{tikzcd}
$$

Finally, we need translate the associativity of $\circ$ and the neutral property of id in terms of commutative diagrams. First observe that, to talk about associativity, we need take the compositions $h \circ (g \circ f)$ and $(h \circ g) \circ f$, which make sense not for any three morphisms, but only for which $d(h) = c(g \circ f)$ and $c(f) = d(h \circ g)$, respectively. So, we don't need look the products

$$\mathrm{Mor}(\mathbf{C}) \times (\mathrm{Mor}(\mathbf{C}) \times \mathrm{Mor}(\mathbf{C})) \quad \text{and} \quad (\mathrm{Mor}(\mathbf{C}) \times \mathrm{Mor}(\mathbf{C})) \times \mathrm{Mor}(\mathbf{C}),$$

but, in each case, to the following pullbacks, which are evidently isomorphics:

$$
\begin{tikzcd}
\mathrm{Pb}(\mathrm{Pb}(d,c),d) \arrow[rr, "\pi_2"] \arrow[d, "\pi_1"'] & & \mathrm{Mor}(\mathbf{C}) \arrow[d, "d"] \\
\mathrm{Pb}(d,c) \arrow[r, "\circ"'] & \mathrm{Mor}(\mathbf{C}) \arrow[r, "c"'] & \mathrm{Ob}(\mathbf{C})
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathrm{Pb}(\mathrm{Pb}(d,c),c) \arrow[rr, "\pi_2"] \arrow[d, "\pi_1"'] & & \mathrm{Mor}(\mathbf{C}) \arrow[d, "c"] \\
\mathrm{Pb}(d,c) \arrow[r, "\circ"'] & \mathrm{Mor}(\mathbf{C}) \arrow[r, "d"'] & \mathrm{Ob}(\mathbf{C})
\end{tikzcd}
$$

With this in mint, the associativity corresponds to the first below diagram, where the segmented arrow and the dotted arrow are ones obtained by universality, as present in the second and in the third diagrams.

$$
\begin{tikzcd}
\mathrm{Pb}(\mathrm{Pb}(d,c),d) \arrow[r, "\simeq"] \arrow[d, dashed] & \mathrm{Pb}(\mathrm{Pb}(d,c),c) \arrow[r, dotted] & \mathrm{Pb}(d,c) \arrow[d, "\circ"] \\
\mathrm{Pb}(d,c) \arrow[rr, "\circ"'] & & \mathrm{Mor}(\mathbf{C})
\end{tikzcd}
$$

$$\begin{tikzcd}
\mathrm{Pb}(\mathrm{Pb}(d,c),d) \arrow[dr, dashed] \arrow[drr, bend left, "(\circ)\circ\pi_1"] \arrow[ddr, bend right, "\pi_2"'] & & \\
& \mathrm{Pb}(d,c) \arrow[r] \arrow[d] & \mathrm{Mor}(\mathbf{C}) \arrow[d, "c"] \\
& \mathrm{Mor}(\mathbf{C}) \arrow[r, "d"'] & \mathrm{Ob}(\mathbf{C})
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathrm{Pb}(\mathrm{Pb}(d,c),c) \arrow[dr, dotted] \arrow[drr, bend left, "\pi_2"] \arrow[ddr, bend right, "(\circ)\circ\pi_1"'] & & \\
& \mathrm{Pb}(d,c) \arrow[r] \arrow[d] & \mathrm{Mor}(\mathbf{C}) \arrow[d, "c"] \\
& \mathrm{Mor}(\mathbf{C}) \arrow[r, "d"'] & \mathrm{Ob}(\mathbf{C})
\end{tikzcd}$$

On the other hand, to describe the neutral property of id, we need take compositions like $id_X \circ f$ and $f \circ id_X$, of mode that we are working in the following pullbacks:

$$\begin{tikzcd}
\mathrm{Pb}(d\circ id, c) \arrow[rr, "\pi_2"] \arrow[d, "\pi_1"'] & & \mathrm{Mor}(\mathbf{C}) \arrow[d, "c"] \\
\mathrm{Ob}(\mathbf{C}) \arrow[r, "id"'] & \mathrm{Mor}(\mathbf{C}) \arrow[r, "d"'] & \mathrm{Ob}(\mathbf{C})
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathrm{Pb}(d, c\circ id) \arrow[rr, "\pi_1"] \arrow[d, "\pi_2"'] & & \mathrm{Mor}(\mathbf{C}) \arrow[d, "d"] \\
\mathrm{Ob}(\mathbf{C}) \arrow[r, "id"'] & \mathrm{Mor}(\mathbf{C}) \arrow[r, "c"'] & \mathrm{Ob}(\mathbf{C})
\end{tikzcd}$$

Therefore, the neutral property of the function id can be described by the commutativity of the below diagram, where like for the associativity of $\circ$, the segmented arrow and the dotted arrow was obtained by universality of pullbacks.

$$\begin{tikzcd}
\mathrm{Pb}(d\circ id, c) \arrow[r, dashed] \arrow[dr, "\pi_1"'] & \mathrm{Pb}(d,c) \arrow[d, "\circ"] & \mathrm{Pb}(d, c\circ id) \arrow[l, dotted] \arrow[dl, "\pi_2"] \\
& \mathrm{Mor}(\mathbf{C}) &
\end{tikzcd}$$

Now, observe that in the above characterization we only use objects, morphisms and pullbacks. So, we can internalize the concept of category in any ambient with pullbacks. More precisely, let $\mathbf{H}$ be a category such that, similar than **Set**, have pullbacks. So, we define a *category internalized in* $\mathbf{H}$ as being an entity composed of two objects $\mathrm{Ob}(\mathbf{C}) \in \mathbf{H}$ and $\mathrm{Mor}(\mathbf{C}) \in \mathbf{H}$, as well four morphisms

$$c, d : \mathrm{Mor}(\mathbf{C}) \to \mathrm{Ob}(\mathbf{C}), \quad id : \mathrm{Ob}(\mathbf{C}) \to \mathrm{Mor}(\mathbf{C}) \quad \text{and} \quad \circ : \mathrm{Pb}(c,d) \to \mathrm{Mor}(\mathbf{C})$$

of $\mathbf{H}$, which satisfy commutativity conditions identical to such present in the previous diagrams. Evidently, an usual category is just a category internalized on **Set**.

We can develop a theory of internalized categories. For example, we can talk about $\mathbf{H}$-*functors* between internalized categories. Indeed, a classical functor between two categories $\mathbf{C}$ and $\mathbf{D}$ can be seen as formed by two functions

$$F : \mathrm{Ob}(\mathbf{C}) \to \mathrm{Ob}(\mathbf{D}) \quad \text{and} \quad F : \mathrm{Mor}(\mathbf{C}) \to \mathrm{Mor}(\mathbf{D})$$

which take any of the previous diagram in $\mathbf{C}$ and return an analogous commutative diagram in $\mathbf{D}$. So, when such categories are internalized on some ambient $\mathbf{H}$, we can define analogously a functor between requiring that the rules F as above be not functions between sets, but morphisms between objects of $\mathbf{H}$. This give a category $\mathbf{Cat_H}$ of categories internalized in $\mathbf{H}$.

Observation. Evidently, the theory of internal categories depends explicitly of the properties of the ambient category. For example, remember that **Set** has the axiom of choice property, which is really important in many proofs. For example, it was used in the construction of products in **Set**, as well in the characterization of equivalences between categories in terms of functors which are weak bijective on objects and bijective on morphisms. Consequently, such characterization cannot be valid in an ambient $\mathbf{H}$ which don't satisfy the axiom of choice. This is the case, for example, when $\mathbf{H} = \mathbf{Top}$.

- anafunctors.
- Morita theory for $\mathbf{Cat_H}$.

Internalized Groupoids

- internalization of groupoids.

Examples

Here we present some examples of internalized categories and grupoids.

Example 3.5 (*monoidal categories*)**.** As was discussed in the example ??, giving a monoid is just the same that giving a category with an unique object. So, any monoid can be regarded as a category $\mathbf{B}M$ with the property that $\mathrm{Ob}(\mathbf{B}M) \in \mathbf{Set}$ is a terminal object. So, we can internalize the classical concept of monoid in any category $\mathbf{H}$ having pullbacks and a terminal object. Particularly, we can internalize it in **Cat**. The result will be a category $\mathbf{BM}$ internalized on **Cat** whose category of objects $\mathrm{Ob}(\mathbf{BM}) \in \mathbf{Cat}$ is a terminal object (that is, is a category with an unique object). So, giving a categorified monoid is just the same that giving the following data:

1. an usual category $\mathbf{C}$ (corresponding to the category of morphisms of $\mathbf{BM}$);
2. functors $c, d : \mathbf{C} \to \mathbf{1}$, where $\mathbf{1}$ is a category with an unique object $*$ (corresponding to the domain and codomain morphisms);
3. functors $\circ : \mathrm{Pb}(c, d) \to \mathbf{C}$ and $id : \mathbf{1} \to \mathbf{C}$ making commutative ?? and ??.

But observe that, being $\mathbf{1}$ terminal, for any category there is an unique functor from such category to $\mathbf{1}$. So, in the above data, $c = d$ is this is unique constant functor which maps any $X \in \mathbf{C}$ into $*$. Consequently, $\mathrm{Pb}(c, d)$ is the whole product $\mathbf{C} \times \mathbf{C}$. Furthermore, the functor $id : \mathbf{1} \to \mathbf{C}$ simply select a distinguished object on $\mathbf{C}$: the image $id(*)$, which will be denoted simply by 1. Thus, the unique nontrivial diagrams are those corresponds to the associativity of $\circ$ and the neutral property of id. So, a categorified monoid (usually called a *strict monoidal category*) is just a category $\mathbf{C}$ together an associative bifunctor $* : \mathbf{C} \times \mathbf{C} \to \mathbf{C}$ and a distinguished neutral object $1 \in \mathbf{C}$, which makes commutative the following diagrams:

$$\begin{tikzcd}
\mathbf{C} \times \mathbf{C} \times \mathbf{C} \arrow[r, "* \times id"] \arrow[d, "id \times *"'] & \mathbf{C} \times \mathbf{C} \arrow[d, "*"] \\
\mathbf{C} \times \mathbf{C} \arrow[r, "*"'] & \mathbf{C}
\end{tikzcd}
\qquad
\begin{tikzcd}
1 \times \mathbf{C} \arrow[r] \arrow[dr, "\pi_1"'] & \mathbf{C} \times \mathbf{C} \arrow[d, "*"] & \mathbf{C} \times 1 \arrow[l] \arrow[dl, "\pi_2"] \\
& \mathbf{C} &
\end{tikzcd}$$

Note that the above diagram is totally identical to ??. Therefore, categorify a monoid M is just the same that internalize the category $\mathbf{B}M$ in $\mathbf{Cat}$. We have a full subcategory $\mathbf{Mon}_{\mathbf{Cat}} \subset \mathbf{Cat}_{\mathbf{Cat}}$ whose objects are monoidal categories and whose morphisms are $\mathbf{Cat}$-functors between the underlying $\mathbf{Cat}$-internalized categories. Such $\mathbf{Cat}$-functors are just usual functors which preserve the products and distinguished objects. So, it are just the strict monoidal functors. Therefore, $\mathbf{Mon}_{\mathbf{Cat}} \simeq \mathrm{Str}\mathbf{MonCat}$.

Example 3.6 (*symmetric monoidal and grupoidal categories*)**.** The situation is totally analogous for commutative monoids and groups. Indeed, a commutative monoid M can be regarded as a category $\mathbf{B}M$ with an unique object with an additional commutative diagram. On the other hand, M is a group precisely when $\mathbf{B}M$ is a groupoid. So, categorify a commutative monoid is just.............

Example 3.7 (*topological groups*)**.** We can also internalize $\mathbf{B}M$ in other categories $\mathbf{H}$ rather than $\mathbf{Cat}$. For the case of monoids, the unique requirement is the existence of pullbacks and of a terminal object. For groups, on the other hand, we also need binary products. So, for example, we can internalize both monoid and groups in $\mathbf{Top}$. The result are just topological monoids and topological groups. So, internalize monoids and groups in $\mathbf{Top}$ is just the same that internalize the category $\mathbf{B}M$ in $\mathbf{Top}$.

Example 3.8 (*Lie groups*)**.** We can try mimetize the above construction saying that Lie groups are just categories $\mathbf{B}M$ internal to $\mathbf{Diff}$. The category $\mathbf{Diff}$ really has terminal objects and binary products, being respectively given by any manifold with and unique object and by $M \times N$ endowed with the product atlas. But as we have seen, it generaly don't have pullbacks. So, in principle *we cannot internalize* $\mathbf{B}M$ *in* $\mathbf{Diff}$. Happens that, if we restrict to transversal maps (particularly, to submersions), then pullback exists. So, we can effectively internalize monoids and groups in $\mathbf{Diff}_{sub}$. The result is a manifold M, endowed with a distinguished point 1, together smmoth submersions $* : M \times M \to M$ and $inv : M \to M$ satisfying the usual conditions. We observe that this coincide with the usual concept of Lie group, because the assumption of smoothness of the product and of the inversion are sufficient to show that they really are submersions. So, Lie groups are just $\mathbf{B}M$ internal to $\mathbf{Diff}_{sub}$.

Example 3.9 (*topological and Lie groupoids*)**.** Any groupoid (not necessarily with an unique object) can be internalized in any category with pullbacks and binary products. The groupoids internalized in $\mathbf{Top}$ and in $\mathbf{Diff}_{sub}$ are respectively called *topological groupoids* and *Lie groupoids.* We give two class of examples of topological groupoids. They can similarly applied to get examples of Lie groupoids.

1. *Cech groupoids.* Let $U_i \hookrightarrow X$ be a cover of a topological space X. To build a groupoid $C(U_i)$, we consider as objects the pairs (x, i), with $x \in U_i$. The unique morphisms are such simply relabel an element $x \in U_i \cap U_j$ as an element of $U_j \cap U_i$. That is, such that $f(x, i) = (x, j)$. The identities are the trivial relabel and the composition are is given by repeated indexing. This is evidently a grupoid. Indeed,
$$\mathrm{Ob}(C(U_i)) = \amalg_i U_i, \quad \mathrm{Mor}(C(U_i)) = \amalg_{i,j} U_i \cap U_j,$$
with domain and codomain function d and c, as well $\circ$ and id, defined as above (observe that the composition is defined on the pullback present below). To turn it a topological groupoid, enough consider the disjoint topology and the subspace topology (for manifolds, to get a Lie groupoid we need consider $U_i \hookrightarrow X$ as being a *good cover*. Examples are given by the covers by domain of coordinate systems);

2. *action groupoids.* Let G be a group which act $\rho : G \times X \to X$ on a set X. So, we can take the quotient X/G to get the space of orbits. In such space, the elements $g \cdot x$, for any $g \in G$, are considered as being an unique. This means that when we work with the class $[x]$ we are leading with any $g \cdot x$ *simultaneously*. We can use this to define a groupoid $X//G$, called the *action groupoid* of G into X, in the following way: the objects are just the elements x. There is a morphism $x \to y$ iff $y = g \cdot x$ for some g. So, the morphisms can be identified with the pairs (g, x). The composition is given by the multiplication in G and the identity of any x is the pair $(1, x)$, where 1 is the neutral element of G. This is really a groupoid with
$$\mathrm{Ob}(X//G) = X, \quad \mathrm{Mor}(X//G) = G \times X$$
and with $c, d, \circ$ and id defined as above. When X and G are topological, then we can turn $X//G$ also topological in an evidently way. When X is smooth and G is a Lie group, in general the pushout X/G *is not a manifold*. But observe that $X//G$ is a Lie groupoid! Indeed, in the set of objects we can take the structure of X and in the set of morphisms the structure of $G \times X$. The problem should occurs in the domain of $\circ$, because it is a pullback and **Diff** don't have arbitrary pullbacks. But in the present situation, such pullback is exactly $G \times G \times X$, which has a product manifold structure.

3. *fundamental groupoid.* We can use the same strategy to codify all fundamental groups of a space (that is, the fundamental groups with different base points) in an unique object! This is the *fundamental groupoid* $\Pi_1(X)$ *of* X. Their objects are just the elements of X. A morphism $x \to y$ is a homotopy class of some path $\gamma : x \to y$, the composition is given by concatenation and the identity of x is the class of the constant path over them. So, more precisely,
$$\mathrm{Ob}(\Pi_1(X)) = X, \quad \mathrm{Ob}(\Pi_1(X)) = [I; X]$$
with $c, d, \circ$ and id as above. This is really a groupoid, because any path has inverse (with respect to concatenation) up to homotopy (observe that $\pi_1(X, x)$ is simply the automorphism group of x). To turn it a topological groupoid,.........

4. *connection groupoid.* https://ncatlab.org/nlab/show/connection+on+a+bundle

Example 3.10 (*smooth groupoids*)**.**

- geometric stacks.

Example 3.11 (*strict 2-groups*)**.**

- talk about non-abelian cohomology and classification of ∞-principal bundles.
- extension of the Seifert-Van Kampen theorem: there is a functor $\Pi_2 : \mathbf{Top}_* \to \mathbf{Grp}_2$ which preserve certain limits. Nonabelian Cohomology. What is the connection with $\pi_2(X,x)$? Is given by the equivalence between 2-groups and crossed modules.
- we can similarly internalize groups in **Gpd**. The result is the same.

Example 3.12 (*Lie 2-group*)**.**

Lie Theory

- if Lie algebras are the infinitesimally versions of the Lie groups, what are the infinitesimally versions of Lie groupoids and of Lie 2-groups? They must be a certain kind of "Lie algebroid" and "Lie 2-algebra". Such notions really exists! They will be discussed later. At this moment we only say that a *strict 2-algebra* is a category internalized in $\mathbf{Vec}_{\mathbb{K}}$ and that a Lie 2-algebra is a particular case of such structures.
- not every Lie algebroid can be integrated. Idea: need to develop a higher categorification theory.

3.2 Enrichment

In the previous section we have been discussed the internalization process, which was used to get more abstract mathematics from classical mathematics. Here we will talk about another of such process: the *enrichment process*. Recall that the internalization process is composed by two steps:

1. characterization of the concept to be internalized (say $\mathcal{P}$) in terms of purely categorical information;
2. definition of an analogous concept in any other category $\mathbf{H}$ (the internalization ambient) in which the categorical information characterizing $\mathcal{P}$ makes sense.

Enrichment is slightly different. Indeed, we can say that *enrichment is a certain kind of partial internalization* (or a certain kind of *internalization of diagrams*). More precisely, enrichment also is composed by two steps and the first of them also is a categorical characterization of $\mathcal{P}$. We observe that part of such categorical information will correspond to commutative diagrams. The second step is then define an analogous of $\mathcal{P}$ in any category $\mathbf{H}$ (here called the *enrichment ambient*) in which an analogous of the commutative diagrams describing $\mathcal{P}$ makes sense. The rest of the structures of $\mathcal{P}$ remains essentially unchanged. Having been done this, we say that $\mathcal{P}$ was *enriched over* $\mathbf{C}$.

The following examples will clasify the ideas:

Example 3.13 (*monoids*)**.** Let is consider the concept of monoid. Recall that it can be fully categorically characterized in terms of a set M, together maps $*: M \times M \to M$ and $1 \to M$ which makes commutative the following diagrams:

$$\begin{tikzcd} X \times X \times X \arrow[r, "* \times id"] \arrow[d, "id \times *"'] & X \times X \arrow[d, "*"] \\ X \times X \arrow[r, "*"'] & X \end{tikzcd} \qquad \begin{tikzcd} 1 \times X \arrow[r] \arrow[dr, "\pi_1"'] & X \times X \arrow[d, "*"] & X \times 1 \arrow[l] \arrow[dl, "\pi_2"] \\ & X & \end{tikzcd}$$

But these diagrams make sense in any monoidal category $\mathbf{C}$! Indeed, enough change $\times$ by $\otimes$ and the set with an unique element 1 by the unity. So, we can talk about *monoid enriched over any strict monoidal category* $\mathbf{C}$ (also called a *monoid object* in $\mathbf{C}$). This is simply an object X, together mappings $X \otimes X \to X$ and $1 \to X$ such that the following diagrams commutes:

$$\begin{tikzcd} X \otimes X \otimes X \arrow[r, "* \otimes id"] \arrow[d, "id \otimes *"'] & X \otimes X \arrow[d, "*"] \\ X \otimes X \arrow[r, "*"'] & X \end{tikzcd} \qquad \begin{tikzcd} 1 \otimes X \arrow[r] \arrow[dr, "\simeq"'] & X \otimes X \arrow[d, "*"] & X \otimes 1 \arrow[l] \arrow[dl, "\simeq"] \\ & X & \end{tikzcd}$$

Evidently, the monoid objects in the cartesian monoidal structure of **Set** are just the classical monoids. Similarly, monoid objects in the cartesian monoidal structures of **Top** and **Diff** are just continuous monoids and differentiable monoids, which was discussed in the examples ?? and ??. But there are monoidal structures which are not cartesian. So, to clarify the ideas, let is give some examples of monoid objects in such kind of entity:

1. *algebras over a ring.* Let is identify the monoid objects in the category of R-modules with the monoidal structure given by the tensor product. By definition, these are R-modules M endowed with R-linear maps $*: M \otimes M \to M$ and $R \to M$ such that the usual "monoidal diagrams" are commutative. But, by the universality of the tensor product,

$$\mathrm{Hom}_R(M \otimes M; M) \simeq \mathrm{Bil}_R(M \times M; M),$$

so that $*$ is nothing more than a bilinear operation on M, which is associative by the commutativity of the first monoidal diagram. On the other hand R is a free R-module of rank one, which is generated by their multiplicative unity 1. So, the given R-linear map $R \to M$ is totally determined by their image on 1 and, thanks to the commutativity of the second monoidal diagram, it behaves as an unity in M with respect to $*$. Therefore, monoids in $\mathbf{Mod}_R$ are just the same that unital associative algebras over a ring R. In particular, unital rings are equivalent to monoids in **AbGrp** (because we have an isomorphism between abelian groups and $\mathbb{Z}$-modules);

2. <u>*monoid objects in cocartesian structure.*</u> Recall that binary coproducts and initial objects in $\mathbf{C}$ defines a cocartesian monoidal structure. A monoid object in it is given by an object X with morphisms $*: X \oplus X \to X$ and $\varnothing \to X$ satisfying the "monoid-like" diagrams. Observe that, being $\varnothing$ an initial object, then there is an unique map $\varnothing \to X$. So, if exists a monoid object structure in a given X then it must has such map as unity. There is a natural multiplication $*: X \oplus X \to X$ given by the codiagonal map ∇, obtained from the universality of binary coproducts. Also by universality, we see that such map is associative (in the sense that it makes the first "monoid-like" diagram commutive). Furthermore, it is the unique map for which the unique $\varnothing \to X$ really is an unity. So, in a cocartesian monoidal structure, any object can be seen as a monoid object in an unique way. The conclusion is that *the notion of monoid object is highly dependent of underlying monoidal structure.*

3. <u>*monoids in monoids.*</u> We have a category of all monoid object in $\mathbf{C}$. Indeed, a morphism between two monoid objects X and X' is simply a morphism $f : X \to X'$ in $\mathbf{C}$ which commutes with the multiplications and with unities. Such category will be denoted by $\mathrm{Mon}(\mathbf{C}, \otimes)$. If $\mathbf{C}$ is *symmetric* then their monoidal structure induces a monoidal structure in $\mathrm{Mon}(\mathbf{C}, \otimes)$, of mode that in such situation we can talk about *monoid object in the category of monoid objects.* Indeed, we first observe that we always have a trivial monoid object structure in $1 \in \mathbf{C}$, whose multiplication $1 \otimes 1 \to 1$ is an equivalence and whose unity $1 \to 1$ is the identity map. Furthermore, the bifunctor $\otimes$ extends to another bifunctor

$$\otimes_M : \mathrm{Mon}(\mathbf{C}, \otimes) \times \mathrm{Mon}(\mathbf{C}, \otimes) \to \mathrm{Mon}(\mathbf{C}, \otimes)$$

which take two monoid objects (say X and Y, with multiplications $*: X \otimes X \to X$ and $*' : Y \otimes Y \to Y$, as well with unities $1 \to X$ and $1 \to Y$) and return a monoid object structure on the product $X \otimes Y$, whose multiplication is given by the map

$$(X \otimes Y) \otimes (X \otimes Y) \xrightarrow{\simeq} (X \otimes X) \otimes (Y \otimes Y) \xrightarrow{* \otimes *'} X \otimes Y$$

and whose unity is given by the map

$$1 \xrightarrow{\simeq} 1 \otimes 1 \longrightarrow X \otimes Y\ ,$$

where the symmetry of $\mathbf{C}$ was used to define the first of such maps. Now, we can check that $\otimes_M$ really defines a monoidal structure in $\mathrm{Mon}(\mathbf{C}, \otimes)$ having the trivial monoid object 1 as unity.

Example 3.14 (*commutative monoids*)**.** Commutative monoids also has a categorical characterization. Indeed, they are just monoids with the additional requirement that the first below diagram is commutative. Such diagram makes sense in any symmetric monoidal category. So, we can talk about *commutative monoid objects* in symmetry monoidal categories, which are characterized by the second below diagram. We have a category of such objects, which will be denoted by ${}_c\mathrm{Mon}(\mathbf{C}, \otimes)$. In the last example we have seen that in the context of symmetric monoidal categories we also can talk about *monoid object in monoid objects.* Happens that such two notions are the same! More precisely, for any symmetric monoidal category $\mathbf{C}$ we have an equivalence

$$\mathrm{Mon}(\mathrm{Mon}(\mathbf{C}, \otimes), \otimes_M) \simeq_c \mathrm{Mon}(\mathbf{C}, \otimes).$$

This is the content of the so called *Eckmann-Hilton argument*, which will be briefly explained now. Details can be seen, for example, in ?? or ??. First recall that a monoid object in $\mathrm{Mon}(\mathbf{C}, \otimes)$ is just a monoid object $(X, *, e)$ in $\mathbf{C}$, where $*$ is the multiplication and e is the unity, together morphisms of monoid objects

$$(X, *, e) \otimes_M (X, *, e) \to (X, *, e) \quad \text{and} \quad (1, \simeq, id_1) \to (X, *, e)$$

making commutative certain "monoid-like" diagrams. So, a monoid object in $\mathrm{Mon}(\mathbf{C}, \otimes)$ is not more than a monoid object $\mathbf{C}$ with additional commutative conditions on certain diagrams. We can see that such additional conditions means exactly that the underlying monoid is commutative.

******** DIAGRAMS**********

Example 3.15 (*comonoids*)**.** We have a partial dual notion of *comonoid.* This is given by a set M endowed with a comultiplication $* : M \to M \times M$ as well with a counit $M \to 1$ making commutative the following (arrows inverted) diagrams:

$$\begin{tikzcd}
X \times X \times X & X \times X \arrow[l, "* \times id"'] \\
X \times X \arrow[u, "id \times *"] & X \arrow[l, "*"] \arrow[u, "*"']
\end{tikzcd}
\qquad
\begin{tikzcd}
1 \times X \arrow[dr, "\pi_1"'] & X \times X \arrow[l] \arrow[r] & X \times 1 \arrow[dl, "\pi_2"] \\
& X \arrow[u, "*"'] &
\end{tikzcd}$$

As for monoids, such dual diagrams also make sense in any strict monoidal category $\mathbf{C}$. Therefore, we can talk about *comonoid objects* in $\mathbf{C}$. These are objects X together morphisms $* : X \to X \otimes X$ and $X \to 1$ satisfying commutative diagrams analogous to the above. Some examples are the following:

1. *comonoids in cartesian structure.* fsdfsdfsdfs

2. *only trivial comonoids.* The last example show that in the cartesian structure there are only trivial comonoids. We observe that there are also non cartesian monoidal structure in which the same holds. With effect, let's consider the cocartesian monoidal structure in **Set**. In it, a comonoid is a set X together maps $X \to X \sqcup X$ and $X \to \varnothing$ satisfying "comonoid-like" diagrams. But there is $X \to \varnothing$ if, and only if, $X = \varnothing$. So, the unique comonoid in such monoidal structure is the trivial;

3. *coalgebras over a ring.* The comonoid objects in $\mathbf{Mod}_R$ with respect to the monoidal structure inhered by the tensor product corresponds to what is called *coalgebras over the ring R.* Particular situation happens when $\mathbb{K}$ is a field and the vector spaces in question are all finite dimensional................

4. *comonoid as monoid in the opposite category.* Recall that **Cat** is complete and cocomplete. So, in particular, it have binary products as well a terminal object, so that it has a cartesian monoidal structure. Therefore, we can talk about monoid object in **Cat**. These are categories $\mathbf{C}$ together multiplication $* : \mathbf{C} \times \mathbf{C} \to \mathbf{C}$ and unity $\mathbf{1} \to \mathbf{C}$ satisfying "monoid-like" diagrams. Therefore, they are just monoidal categories! That is,

Mon($\mathbf{Cat}, \times$) $\simeq$ Str($\mathbf{MonCat}$). Now, as rapidly can be checked, if a functor between monoidal categories is monoidal, then it preserve monoid objects. We have the opposite functor $(-)^{op} : \mathbf{Cat} \to \mathbf{Cat}$ for that $(\mathbf{C} \times \mathbf{D})^{op} \simeq \mathbf{C} \times \mathbf{D}$ and $\mathbf{1}^{op} \simeq \mathbf{1}$. So, it is monoidal and, therefore, preserve monoid object structure. This means that, if $\mathbf{C}$ is monoidal, then their monoidal structure induces another in $\mathbf{C}^{op}$. Therefore, we can talk about monoids in $\mathbf{C}^{op}$, which are just comonoids in $\mathbf{C}$. Now, we could think that such identification extends to an equivalence

$$\mathrm{CoMon}(\mathbf{C}, \otimes) \simeq \mathrm{Mon}(\mathbf{C}^{op}; \otimes).$$

However, this is **not** the case. Indeed, a morphism between monoids in $\mathbf{C}^{op}$ and a morphism between comonoids in $\mathbf{C}$ are respectively maps of the form

$$(X, *, e) \otimes_M (X, *, e) \to (X, *, e) \quad \text{and} \quad (X, *, e) \to (X, *, e) \otimes_M (X, *, e),$$

where $* : X \to X \otimes X$ is the comultiplication and $e : X \to 1$. So, the correct equivalence is

$$\mathrm{CoMon}(\mathbf{C}, \otimes) \simeq \mathrm{Mon}(\mathbf{C}^{op}; \otimes)^{op}.$$

5. *dual Eckmann-Hilton argument.* As consequence of the above characterization, we can see that the Eckmann-Hilton argument also holds for comonoids. More precisely, we have a category CoMon($\mathbf{C}, \otimes$) of comonoid objects in any monoidal category $\mathbf{C}$. Furthermore, if $\mathbf{C}$ is symmetric, then it induces a natural monoidal structure in CoMon($\mathbf{C}, \otimes$) and then we can talk about comonoid objects in the category of comonoid objects. On the other hand, we can also talk about commutative comonoids, which defines a category ${}_c$CoMon($\mathbf{C}, \otimes$). We asserts that

$$\mathrm{CoMon}(\mathrm{CoMon}(\mathbf{C}, \otimes), \otimes_M) \simeq {}_c\mathrm{CoMon}(\mathbf{C}, \otimes).$$

Indeed, by the previous characterization, we have that

$$\begin{aligned} \mathrm{CoMon}(\mathrm{CoMon}(\mathbf{C}, \otimes), \otimes_M) &\simeq \mathrm{CoMon}(\mathrm{Mon}(\mathbf{C}^{op}; \otimes)^{op}) \\ &\simeq \mathrm{Mon}((\mathrm{Mon}(\mathbf{C}^{op}; \otimes)^{op})^{op})^{op} \\ &\simeq \mathrm{Mon}(\mathrm{Mon}(\mathbf{C}^{op}; \otimes))^{op} \end{aligned}$$

and

$${}_c\mathrm{CoMon}(\mathbf{C}, \otimes) \simeq_c \mathrm{Mon}(\mathbf{C}^{op}; \otimes)^{op}.$$

So, what we are asserting it that

$$\mathrm{Mon}(\mathrm{Mon}(\mathbf{C}^{op}; \otimes))^{op} \simeq_c \mathrm{Mon}(\mathbf{C}^{op}; \otimes)^{op}.$$

Therefore, the Eckmann-Hilton for monoids (applied to $\mathbf{C}^{op}$) and the duality principle imply the required.

Example 3.16 (*bimonoids*)**.** We have a partial dual notion of *comonoid*. This is given by a set M endowed with a comultiplication $* : M \to M \times M$ as well with a counit $M \to 1$ making commutative the following (arrows inverted) diagrams:

$$
\begin{tikzcd}
X \times X \times X & X \times X \arrow[l, "* \times id"'] \\
X \times X \arrow[u, "id \times *"] & X \arrow[l, "*"] \arrow[u, "*"']
\end{tikzcd}
\qquad
\begin{tikzcd}
1 \times X \arrow[dr, "\pi_1"'] & X \times X \arrow[l] \arrow[r] & X \times 1 \arrow[dl, "\pi_2"] \\
 & X \arrow[u, "*"'] &
\end{tikzcd}
$$

Example 3.17 (*Hopf monoids*).

Example 3.18 (*Frobenius monoids*).

Microcosm

Recall that strict monoidal categories was obtained categorifying monoids (which is a concept inside **Set**) in **Cat**. When we replace a concept by other more abstract, is expect to be possible recover (or localize) the oldest concept as a particular case of the newest. Therefore, we expect be possible recover the classical concept of monoid in terms of the language of categories with an unique objects and in terms of the language of monoidal categories.

Therefore, *a classical monoid (which is internal to* **Set***) can be described in the language of monoidal categories as been a monoid object in the monoidal category of sets.* In diagrammatic terms, the below sequence produces the identity map:

We observe that this idea in principle can be applied by many other concepts. Indeed, the we expect that, in the following recipe, if the third step is true, them the forth is true as well:

1. consider a classical concept. Categorifying it we get a categorical concept;

2. suppose that the categorical concept is a category endowed with additional structures and that **Set** is an example of them;

3. *the classical concept can be internalized in any categorical version of them.* Particularly, it can be internalized in **Set**;

4. internalizing the classical concept in **Set** we get the classical concept again.

Observe that the second condition holds, for example, when the concept to be categorified admits an "one object" characterization. This is the case of many algebraic structures, like monoids, groups, rings and their commutative versions. So, in this context, the third step say that *certain algebraic structures can be internalized in any categorical version of them.* This is the so called *microcosmic principle*, term which was coined by Baez and Dolan in [**?**].

We can see that, not only for monoids, but also for groups, rings, commutative monoids, etc, this works very well. A more abstract formulation needs higher categorical language and, as will be discussed later, this means that the term "principle" at "micromosm principle" may have different interpretations depending of the axiomatic language used. For instance, we can have a "microcosmic hypothesis" or else a "microcosmic theorem". Proofs of the "microcosmic theorem" can be seen, for example, in [**?**, **?**].

Enriched Categories

Recall that a semicategory $\mathbf{C}$ is composed by a collection of objects $\mathrm{Ob}(\mathbf{C})$ and, for any pair of such objects another collection of morphisms $\mathrm{Mor}_{\mathbf{C}}(X;Y)$, which are linked by certain composition maps

$$\circ : \mathrm{Mor}_{\mathbf{C}}(X;Y) \times \mathrm{Mor}_{\mathbf{C}}(Y;Z) \to \mathrm{Mor}_{\mathbf{C}}(X;Z)$$

satisfying associative conditions, in the sense that the following diagram commutes:

$$\begin{tikzcd}
\mathrm{Mor}_{\mathbf{C}}(X;Y) \times \mathrm{Mor}_{\mathbf{C}}(Y;Z) \times \mathrm{Mor}_{\mathbf{C}}(Z;W) \arrow[r, "id\times\circ"] \arrow[d, "\circ\times id"'] & \mathrm{Mor}_{\mathbf{C}}(X;Y) \times \mathrm{Mor}_{\mathbf{C}}(Y;W) \arrow[d, "\circ"] \\
\mathrm{Mor}_{\mathbf{C}}(X;Z) \times \mathrm{Mor}_{\mathbf{C}}(Z;W) \arrow[r, "\circ"'] & \mathrm{Mor}_{\mathbf{C}}(X;W)
\end{tikzcd}$$

But such diagrams makes sense in any monoidal category! So, we can talk about *semicategories enriched over a monoidal category* $(\mathbf{H}, \otimes, 1)$. Indeed, these are entities obtained replacing $\times$ by $\otimes$ and functions by morphisms of $\mathbf{H}$. More precisely, a semicategory enriched over $\mathbf{H}$ is composed by a collection of objects $\mathrm{Ob}(\mathbf{C})$ and, for any pair $X, Y \in \mathrm{Ob}(\mathbf{C})$, an object $\mathbf{H}(X,Y)$ of $\mathbf{H}$, which are linked by morphisms

$$\circ : \mathbf{H}(X;Y) \otimes \mathbf{H}(Y;Z) \to \mathbf{H}(X;Z)$$

making commutive the following diagram:

$$\begin{tikzcd}
\mathbf{H}(X;Y) \otimes \mathbf{H}(Y;Z) \otimes \mathbf{H}(Z;W) \arrow[r, "id\otimes\circ"] \arrow[d, "\circ\otimes id"'] & \mathbf{H}(X;Y) \otimes \mathbf{H}(Y;W) \arrow[d, "\circ"] \\
\mathbf{H}(X;Z) \otimes \mathbf{H}(Z;W) \arrow[r, "\circ"'] & \mathbf{H}(X;W)
\end{tikzcd} .$$

When the semicategory $\mathbf{C}$ to be enriched really is a category, then it also has unities. This means that for any object X there is a morphism $id_X : X \to X$, or equivalently a function $id_X : * \to \mathrm{Mor}_{\mathbf{C}}(X,X)$, making commutive the first below diagram. Such diagram makes sense in a monoidal category, so that *not only semicategories, but also categories can be enriched in any monoidal categories.* Indeed, an enriched category over $\mathbf{H}$ is an enriched semicategory in which any $X \in \mathrm{Ob}(X)$ has a morphism $id_X : 1 \to \mathbf{H}(X,X)$ making commutative the second diagram below.

******* DIAGRAM **********

Given two $\mathbf{H}$-enriched categories $\mathbf{C}$ and $\mathbf{D}$, we have a natural notion of $\mathbf{H}$-*enriched functors* between them. These are composed by a function on objects and morphisms of $\mathbf{H}$ between the hom-objects, like

$$F : \mathrm{Ob}(\mathbf{C}) \to \mathrm{Ob}(\mathbf{D}) \quad \text{and} \quad F : \mathbf{H}_{\mathbf{C}}(\mathrm{X}, \mathrm{Y}) \to \mathbf{H}_{\mathbf{D}}(\mathrm{F}(\mathrm{X}), \mathrm{F}(\mathrm{Y})),$$

which preserve the composition morphisms and the unities in the sense that the following diagrams are commutative:

********** DIAGRAM *********

So, we have a category **Cat**(**H**) of **H**-enriched categories and **H**-enriched functors between them. The area of math corresponding to the study of such king of categories is *enriched category theory*. Evidently, an usual category is simply a **Set**-enriched category, so that the study of **Cat**(**Set**) corresponds to usual category theory what was developted in the previous chapter. Many concepts and results of classical category theory admits natural extensions to the enriched case. A classical text about this is [**?**]. For instance, we will see that we have a generalized notions of limit, Kan extensions and ends in any enriched category, so that, like in the classical case, limits are particular Kan extensions which are totally determined by ends.

Examples

There are some particularly interesting examples of enriched categories. They will be briefly discussed in this subsection.

Example 3.19 (*abelian categories*)**.** A category enriched over $(\mathbf{Mod}_R, \otimes)$ corresponds to what in the literature is usually called *R-linear category*. This is simply a category in which their sets of morphisms are endowed with a structure of R-modules in such way that the compositions are bilinear. Special R-linear categories are those which has biproducts, a null object as well kernels and cokernels (this happens, for example, when it has finite limits and colimits). Such categories are called *abelian* and are an abstract context in which we can study Homological Algebra. Indeed,..

Example 3.20 (*strict 2-categories*)**.**

Example 3.21 (*monoid objects and monoidal categories*)**.** Exactly as a monoid is an usual category with an unique object, a monoid object in a monoidal category **H** is just a **H**-enriched category with an unique object. Particularly, the (strict) monoidal categories corresponds to the strict 2-categories with an unique object.

- observation: the previous example clarify a relation between enrichment and internalization: strict monoidal categories can be identified as 2-categories with an unique object. This can be seen as a version of the microcosm principle. This relation will be extended in the next section.

Weighed Limits and Ends

- in enriched categories we can define ends and coends in a similar way. That is, we can internalize ends and coends (and particularly limits and colimis) in any enriched category.

3.3 Highering

- higher categorification
- strict n-categories

- strict ∞-categories
- higher microcosmic principle
- stabilization hypothesis of Baez and Dolan (as higher Eckman-Hilton)
- ∞-ends and ∞-coends as colimits of enriched ends and coeds.

Chapter 4

Weakening

- the necessity of weaken (for example, to see **Cob**$(n+1)$ as a categorification of **Cob**(n). To see the homotopy category of **Top** as a 2-category) and to give good examples of monoidal categories.

4.1 Coherence

- weaken first and after

4.2 Presentations

ω-Nerve

Definitions

4.3 Theory

- ∞-limits and colimits
- internal concepts.

Chapter 5

Homotopy

5.1 Approximation

- every homotopical category is the homotopy category of some $(\infty, 1)$-category (slogan demonstrado na monografia)
- homotopy limits: local definition and global definition. Utility of model categories: equivalence between these two notions of homotopy limits.

5.2 Models

Examples

- problems with define a model on **Cat**. Motivation for anafunctors.
- Strom model on the $(\infty, 1)$-category **Top**;
- Quillen model on the $(\infty, 1)$-category **Top**.
- Injective and Projective models on the $(\infty, 1)$-categories $\mathbf{Ch_{Ab}}$ and $\mathbf{CCh_{Ab}}$.
- Injective and Projective models on $\mathbf{Cat}_\omega$.

Quillen Equivalences

- homotopy hypothesis

5.3 Utility

- The main utility of model introduced on $(\infty, 1)$-categories is the possibility to translate many higher categorical information as usual 1-categorical information added of certain conditions.

5.4 Stabilization

- looping and delooping.
- category of spectrum.

Chapter 6

∞-Toposes

6.1 Structure

6.2 Presentations

6.3 Properties

Chapter 7

Invariants

7.1 Homotopy

We are looking to homotopical invariants of a $(\infty, 1)$-category $\mathbf{C}$. As we have said, invariants are generally obtained from functors, of mode that we are looking to functors

$$F : \mathrm{Ho}(\mathbf{C}) \to \mathbf{Alg}$$

which take values in some algebraic category. The most simple functors on any category are the hom-functors. So, for any given A, we can consider

$$[A; -]_{\mathbf{C}} = \mathrm{Mor}_{\mathrm{Ho}\mathbf{C}}(A; -) = \pi_0(\mathrm{Mor}_{\mathbf{C}}(A; -)).$$

For a general A, such functors take values only in the category of sets and, therefore, they not define rich invariants. The question is: *there is some A for that the correspondent hom-functor take values in some algebraic category?* Remember that........

Examples

- classical groups, simplicial groups and their equivalence between homotopy hypothesis.

Fibration Sequence

Whitehead Tower

7.2 Cohomology

Cohomology is the complete dual version of homotopy.

- dual to homotopy

Abelian Cohomology

- example: De Rham cohomology,

7.3 Examples

Appendix A

Fubini

Here we will give an outline of the proof of the Fubini's theorem for ends. The idea for coends will be totally dual. So, let

$$E : (\mathbf{C} \times \mathbf{D})^{op} \times (\mathbf{C} \times \mathbf{D}) \to \mathbf{D}'$$

be a functor. We will show that the universality of the end $\int_{\mathbf{C}\times\mathbf{D}} E$ induces a natural map

$$u : \int_{\mathbf{D}} \int_{\mathbf{C}} E \to \int_{\mathbf{C}\times\mathbf{D}} E.$$

The universality of $\int_{\mathbf{D}} \int_{\mathbf{C}} E$ will induce, similarly, a map in the opposite direction. By construction, such maps will be one the inverse of the another, ending the proof.

To build u, let consider the below diagram. Their basis is simply the equalizer diagram which defines the consecutive end $\int_{\mathbf{D}} \int_{\mathbf{C}} E$. To simplify the notation, we write

$$E^g = E(X, X, d(g), d(g)), \quad E_g = E(X, X, c(g), c(g)) \quad \text{and} \quad E^g_g = (X, X, d(g), c(g)).$$

The top, on the other hand, is the equalizer diagram defining the double end $\int_{\mathbf{C}\times\mathbf{D}} E$, having been made the identifications

$$\Pi_{(X,Z)} E \simeq \Pi_Z \Pi_X E \quad \text{and} \quad \Pi_{(f,g)} E \simeq \Pi_g \Pi_f E,$$

and having been used the simplified notation

$$E^{fg} = E(d(f), c(f), d(g), d(g)) = E(d(f, g), c(f, g)), \quad \text{etc.}$$

The vertical arrows of the cube are provided by the equalizer which defined $\int_{\mathbf{C}} E$. Now, the compositions of the tilde arrows and the pointed arrows produce maps

$$\int_Z \int_X E \to E^{fg} \quad \text{and} \quad \int_Z \int_X E \to E_{fg},$$

of mode that, by the universality of products, they induces the curved arrow. Now, how each face commuted, we have $a \circ c = b \circ c$. So, by the universality of equalizers, there is the double dotted arrow, corresponding to the required map u.

$$
\begin{tikzcd}
 & & & E^{gf} \arrow[rrr] & & & E_{gf} \\
\int_{(X,Z)} E \arrow[urrr] \arrow[rr] \arrow[dr] & & \Pi_Z \Pi_X E \arrow[ur] \arrow[dl] \arrow[rrr, shift left, "a"] \arrow[rrr, shift right, "b"'] & & & \Pi_g \Pi_f E \arrow[ur] \arrow[dl] & \\
 & E_{gf} \arrow[rrr] & & & E^{gf}_{gf} & & \\
 & & & \int_X E^g \arrow[uuu, dotted] \arrow[rrr, dashed] & & & \int_X E^g_g \arrow[uuu] \\
\int_Z \int_X E \arrow[uuu, dotted, Rightarrow, "\exists!"] \arrow[urrr, dotted] \arrow[rr, dashed] \arrow[dr, squiggly] \arrow[uuurr, dashed, bend left, "c"'] & & \Pi_Z \int_X E \arrow[ur, dashed] \arrow[dl, dashed] \arrow[rrr, dashed, shift left] \arrow[rrr, dashed, shift right] & & & \Pi_g \int_X E^g_g \arrow[ur] \arrow[dl] & \\
 & \int_X E_g \arrow[uuu, squiggly] \arrow[rrr] & & & \int_X E^g_g \arrow[uuu] & &
\end{tikzcd}
$$